

A Hybrid Framework for Phishing Email Detection and Secure Communication Using AI and Blockchain

Dissertation submitted in fulfilment of the requirements for the Degree of

**MASTER OF TECHNOLOGY in
INFORMATION SECURITY AND CYBER FORENSICS**

COMPUTER SCIENCE AND ENGINEERING

By

Kadiatou Diallo

Registration number

12400058

Supervisor

Dr. Puneet Thapar

Associate Professor

**Startup, Innovation and Entrepreneurship Committee Coordinator School
of Computing and Artificial Intelligence**



School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab (India)

April 2026



TOPIC APPROVAL PERFORMANCE

School of Computer Science and Engineering (SCSE)

Program: P17AD M.Tech. (Information Security and Cyber Forensics) [Full Time]

COURSE CODE: CSE587

REGULAR/BACKLOG: Regular

GROUP NUMBER: CSERGD0626

Supervisor Name: Dr. Puneet Thapar

UID: 28156

Designation: Associate Professor

Qualification:

Research Experience:

SR.NO.	NAME OF STUDENT	Prov. Regd. No.	BATCH	SECTION	CONTACT NUMBER
1	Kadiatou Diallo	12400058	2024	K24IS	7901846558

SPECIALIZATION AREA: Program Methodology and Design

Supervisor Signature:

PROPOSED TOPIC: A Hybrid Framework for Phishing Email Detection and Secure Communication Using AI and Blockchain

Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	Parameter	Rating (out of 10)
1	Project Novelty: Potential of the project to create new knowledge	6.90
2	Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.	6.90
3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	6.90
4	Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.	7.00
5	Social Applicability: Project work intends to solve a practical problem.	7.00
6	Future Scope: Project has potential to become basis of future research work, publication or patent.	7.00
PAC Committee Members		
PAC Member (HOD/Chairperson) Name: Gaurav Pushkarna	UID: 11057	Recommended (Y/N): Yes
PAC Member (Allied) Name: Dr. Harpreet Kaur	UID: 27633	Recommended (Y/N): Yes
PAC Member 3 Name: Dr. Gauri Mathur	UID: 11400	Recommended (Y/N): Yes

Final Topic Approved by PAC: A Hybrid Framework for Phishing Email Detection and Secure Communication Using AI and Blockchain

Overall Remarks: Approved

PAC CHAIRPERSON Name: 17442 Dr. Arun Malik

Approval Date: 23 Apr 2025

4/24/2025 2:19:28 PM

Abstract

Phishing continues to be one of the most common and dangerous forms of cybersecurity threats, especially when using email-based communication platforms for stealing valuable information such as passwords, bank details, and even identities. Current phishing detection methods have become insufficient due to the sophistication and constantly developing strategies behind the crime. While machine learning algorithms have been successfully used to improve phishing detection, the methods have been proven to have several shortcomings such as low robustness and inability to verify collected data. Therefore, the main purpose of this paper is to develop a framework, called CrypPhish AI, which combines machine learning with blockchain to improve phishing detection and provide security during communication. Specifically, the algorithm uses different machine learning models, such as Logistic Regression, Random Forest, Support Vector Machine (SVM), and LSTM to identify suspicious emails through analysis of features in the header, message body, URLs, and metadata. The experiment showed that SVM has demonstrated the best results, with up to 99% accuracy. To maintain the integrity and security of data, a blockchain verification module is used where validated data about the email are stored in form of hashes using SHA-256 algorithm. The hashes are stored in a secure manner with the help of smart contracts in an Ethereum based network. The benefits offered by this technique include the features of immutability, transparency, non-repudiation, and traceability, in addition to maintaining privacy as hashing of data ensures the use of only hashes on the blockchain. Selective storage is employed in which only valid emails are stored in the blockchain, thereby saving the computing resources. It can be seen from the above discussion that AI and blockchain combination offer a very efficient solution for detecting phishing attacks.

DECLARATION STATEMENT

I hereby declare that the research work reported in the dissertation/dissertation proposal entitled “**A Hybrid Framework for Phishing Email Detection and Secure Communication Using AI and Blockchain** ” in partial fulfilment of the requirement for the award of Degree for Master of Technology in Computer Science and Engineering at Lovely Professional University, Phagwara, Punjab is an authentic work carried out under supervision of my research supervisor Mr./Mrs. Research Guide’s Name. I have not submitted this work elsewhere for any degree or diploma.

I understand that the work presented herewith is in direct compliance with Lovely Professional University’s Policy on plagiarism, intellectual property rights, and highest standards of moral and ethical conduct. Therefore, to the best of my knowledge, the content of this dissertation represents authentic and honest research effort conducted, in its entirety, by me. I am fully responsible for the contents of my dissertation work.

Signature of Candidate

Kadiatou Diallo

R.No 12400058

SUPERVISOR'S CERTIFICATE

This is to certify that the work reported in the M. Tech Dissertation/dissertation proposal entitled **“A Hybrid Framework for Phishing Email Detection and Secure Communication Using Ai and Blockchain”**, submitted by **Kadiatou Diallo** at **Lovely Professional University, Phagwara, India** is a bonafide record of his / her original work carried out under my supervision. This work has not been submitted elsewhere for any other degree.

Signature of Supervisor

Dr. Puneet Thapar

Date:

ACKNOWLEDGMENT

I would like to express my sincere gratitude to my supervisor, Dr. Puneet Thapar, for his continuous guidance, valuable suggestions, and unwavering support throughout the completion of this research work. His expertise and encouragement played a crucial role in shaping this study and improving its quality.

I am also thankful to the faculty members of the School of Computer Science and Engineering, Lovely Professional University, for providing the necessary academic environment and resources that contributed to the successful completion of this research.

I would like to extend my appreciation to my friends and colleagues for their support and motivation. I would especially like to thank my friend Richard Sarpong, whose assistance, continuous encouragement, and valuable contributions significantly supported the completion of this research work.

I also acknowledge the use of publicly available datasets and open-source tools that supported the experimental implementation of this study.

Finally, I express my deepest gratitude to my family for their constant encouragement, patience, and moral support, which have been the foundation of my academic journey.

TABLE OF CONTENTS

Contents

CHAPTER 1	1
INTRODUCTION	1
1.1 SCOPE OF PROJECT	2
1.2 RESEARCH QUESTIONS	3
CHAPTER 2	4
RELATED WORK	4
2.1 MACHINE LEARNING BASED	4
2.2 CRYPTOGRAPHIC DEFENSES & SPECIALIZED SOLUTIONS	6
2.3 BLOCKCHAIN-BASED APPROACHES	7
CHAPTER 3	9
PRESENT WORK	9
3.1 PROBLEM FORMULATION	9
3.2 OBJECTIVES OF THE STUDY	10
3.3 RESEARCH METHODOLOGY	11
CHAPTER 4	17
RESULTS AND DISCUSSION	17
4.1 EXPERIMENTAL SETUP	17
4.3 RESULTS OF MACHINE LEARNING MODELS	20
4.4 DISCUSSION OF RESULTS	26
4.6 DISCUSSION OF BLOCKCHAIN RESULTS	29
4.7 WEB INTERFACE RESULTS	30
4.9 EXAMPLE OF PHISHING DETECTION OUTPUT	34
4.10 OVERALL DISCUSSION AND CONCLUSION OF RESULTS	36
4.11 COMPARISON WITH EXISTING TECHNIQUES	36
4.12 DISCUSSION OF COMPARISON	37
CHAPTER 5	38
FUTURE SCOPE	38
CHAPTER 6	39
CONCLUSION	39

REFERENCES.....	41
PUBLICATIONS.....	45

LIST OF ACRONYMS

Acronym	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
NLP	Natural Language Processing
TF-IDF	Term Frequency–Inverse Document Frequency
SPF	Sender Policy Framework
DKIM	DomainKeys Identified Mail
DMARC	Domain-based Message Authentication, Reporting and Conformance
SHA-256	Secure Hash Algorithm 256-bit
ETH	Ethereum
Tx	Transaction
PGP	Pretty Good Privacy
ROC	Receiver Operating Characteristic
BERT	Bidirectional Encoder Representations from Transformers
ICPCSN	International Conference on Pervasive Computing and Social Networking
ICCCA	International Conference on Computing, Communication and Automation

LIST OF FIGURES

FIGURE NO.	FIGURE DESCRIPTION	PAGE NO.
Figure 1	System Integration and Architecture Flowchart	14
Figure 2	Text Conversion using TF-IDF vectorization	15
Figure 3	Content of the Nigerian_Fraud Emails Dataset	18
Figure 4	Content of the SpamAssasin Email Dataset	18
Figure 5	Cleaned Dataset	19
Figure 6	Class Distribution for Safe and Phishing Emails	19
Figure 7	The Performance Evaluation Metrics for Logistic Regression	21
Figure 8	Confusion Matrix for Logistic Regression	21
Figure 9	ROC Curve for Logistic Regression	22
Figure 10	The Performance Evaluation Metrics for Random Forest	22
Figure 11	Confusion Matrix for Random Forest	22
	Roc Curve for Random Forest	23
Figure 12	The Performance Evaluation Metrics for SVM	23
Figure 13	Confusion Matrix for SVM	23
	ROC Curve for SVM	24
Figure 14	The Performance Evaluation Metrics for LSTM	24
Figure 15	Confusion Matrix for LSTM	24
	ROC Curve for LSTM	25
Figure 16	Bar Graph Showing the various ML models Performance Comparison	25
Figure 17	Email Classification and Storage on Blockchain	28
Figure 18		
Figure 19		

Figure 20		
Figure 21	Ganache Application Platform	28
Figure 22	Ganache Interface Showing Blockchain Blocks and Transactions Generated by the Proposed System	29
Figure 23	Web Application Interface for the system	30
	Various features of the Web Application Interface	31
Figure 24	Highlight of the Result Analyse feature of the Web Application	31
Figure 25	Highlight of the Blockchain History feature	32
Figure 27	Highlight of the Verify Email feature of the Web Application	32
Figure 28	Email Sample being tested on the Web Application	33
Figure 29	Email ready to be analysed	33
Figure 30	Example Output of CrypPhish AI System Showing Safe Email Classification and Blockchain Verification	34
Figure 31	Example Output Showing Phishing Email Detection with High Confidence	35

LIST OF TABLES

TABLE NO.	TABLE DESCRIPTION	PAGE NO.
Table I	Tools and Technology Used	16
Table II	Machine Learning Performance Comparison	26
Table III	Comparison of Existing Methods with Proposed System	37

CHAPTER 1

INTRODUCTION

Phishing cyber-attacks represent one of the commonest and most serious threats in today's digital world. Cyber criminals have continued to use fraudulent methods that involve the exploitation of security weaknesses in order to steal personal information as an increasing number of people, corporate institutions, and governments have come to rely on digital communication channels for their personal information and transactions. Usually, phishing represents a deceitful email or web site message posing as legitimate sources in a bid to get personal information like passwords, bank accounts, and identity numbers from victims. Phishing represents a constant menace in the current cybersecurity environment due to the increased attack surface brought about by increased internet connectivity.

The impacts of phishing are profound and widespread. While the firm may suffer from damage to its reputation, disruption to its operations, and lawsuits, individuals can lose money, be stripped of their identities, and suffer invasions of their privacy. According to studies that discuss cybersecurity, phishing is one of the most widely used means of entry into catastrophic events in cyberspace, such as ransomware attacks and data breaches.

Through time, various means of detection were used in order to counter this increasing problem. The traditional methods include signature matching techniques, rules, and blacklisting systems. Although these techniques have the capability of detecting previously known phishing attacks, they lack the ability to detect newly formed and emerging phishing activities. This was the reason why various interesting methods were employed in order to counter the threat posed by phishing, including machine learning technology that has the capability of detecting any suspicious activity based on the information contained in huge databases.

However, there are several demerits associated with the machine learning systems despite their efficiencies. To evade being caught, the attackers usually come up with new strategies that lead to problems like false positive errors, false negative errors, and model accuracies. The efficiency of the centralized detection systems may be influenced by certain factors like data manipulation. These demerits show the need for more efficient methods that will enable accurate detections of attacks while ensuring the accuracy of detection information.

Another technology which has become popular because of its ability to resolve the problems stated above is blockchain technology. The features provided by blockchain technology such as decentralization, transparency, immutability, and tamper-resistance make the cybersecurity

system reliable. Blockchain technology can make sure that the data is not altered without any reason and provide an effective way to authenticate data for distributed systems. For these reasons, combining blockchain technology with smart detection methods has become an area of increasing interest.

In this regard, this research seeks to investigate the potential of using cutting-edge technologies in phishing detection, particularly considering how a safe framework can be used along with machine learning to enhance its efficiency and accuracy. Through this study, one can make significant contributions towards designing protective measures for phishing attacks by introducing novel ideas and evaluating existing methods. In order to mitigate the challenges posed by cybersecurity issues in today's highly inter-connected world, there is a need to leverage the power of intelligent analysis in combination with robust infrastructure.

1.1 SCOPE OF PROJECT

The main purpose of this study is to develop a system for the detection of phishing email attacks through the application of intelligent learning techniques. In the modern world, phishing is still considered one of the most frequent security risks, especially when it comes to online messaging. The main goal of this research is to develop a detection tool that will be able to recognize phishing emails with high accuracy without producing any false alarms.

This study focuses exclusively on the scope of phishing attacks made using electronic mail communication. It entails analysis of different aspects of electronic mail information like email headers, email body contents, URLs, attachment files, among others. They undergo feature extraction processes, which help detect characteristics that are common to phishing attacks. Machine learning algorithms are deployed to compare various options and determine their effectiveness for classifying phishing attacks, spam messages, and regular electronic mails. This work does not explore other phishing types that include smishing, vishing, among others.

Apart from detection, this study utilizes blockchain technology to make the system more secure and reliable. The blockchain scheme has been implemented to store authenticated emails through cryptographic hashing (SHA-256). This makes the information immutable, traceable, and unalterable. The blockchain part serves as a layer that adds up to the detection process. This makes the detection layer more reliable and secure. This way, only genuine emails will be stored, thus making the system highly trustworthy and efficient in terms of storage and user privacy.

In addition, this research has a certain scope, and that is related only to designing, implementing, and experimentally evaluating a prototype system based on publicly accessible data sets. All experiments have been performed in a well-controlled manner to evaluate the performance

of the models and the functioning of the system. This research is not about the large-scale deployment, live integration with real-world email servers, and industrial implementation. Nevertheless, the framework presented can be considered as a good basis for further research.

1.2 RESEARCH QUESTIONS

Several research questions that aim to explore the effectiveness of intelligent and secure phishing detection approaches form the foundation for this research. These research questions help to set the objectives for the research.

1. To what extent can email communication systems identify phishing attacks through machine learning approaches?
2. What are the limitations of existing and traditional phishing detection approaches in modern-day cybersecurity environments?
3. Can the security, integrity, and reliability of phishing detection systems be enhanced through the application of blockchain technology?
4. What factors influence the performance and accuracy of automated phishing detection models?
5. To what extent can a blockchain and machine learning hybrid model reduce false positives and false negatives as compared to traditional approaches?

Collectively, these research questions address the challenges of identifying aggressive activity within an increasingly complex cyber environment. Through the provision of answers to these research questions, the research study hopes to contribute to the development of more reliable, secure, and efficient anti-phishing defences.

CHAPTER 2

RELATED WORK

On the other hand, a number of artificial intelligence and machine learning techniques have been developed and put out for detecting phishing emails and providing dynamic and data-driven solutions.

2.1 MACHINE LEARNING BASED

Khan and associates [1] have applied classifiers such as Random Forest, SVM, and Logistic Regression for detecting phishing trends, but they have not been compared with dynamic adversarial CNNs with EGSO and VAE upgrades and have also suffered from generality issues because of using tailored datasets without cross-validation.

Lim et al [2] proposed an explainable AI and ML-based framework called EXPLICATE, which showed promising results for phishing detection. However, there is a need for cross-validation of datasets and evaluation of its long-term performance, as it is not clear if it is robust and effective against different kinds of threats.

Chinta et al [3] have suggested the BERT and LSTM hybrid model, which showed 99.55% accuracy on phishing data sets. However, the scalability of this model has not been proven, as the data sets are limited, the computation cost is high, and there is no validation of the model on realtime phishing scenarios.

Asani et al [4] have proposed a phishing detection framework based on the combination of the Gaussian Latent Variable Model for feature refinement and the Support Vector Classifier for classification. The proposed model has shown promise, though it failed to incorporate the benchmarking, versatility, and flexibility required for the validation of the model.

Sahit et al [5] presented a comprehensive phishing detection framework using a combination of machine learning models, advanced feature engineering, and natural language processing for greater flexibility and real-time detection. Despite the advantages, the framework also needs multilingual support and validation for effective deployment.

Mahalakshmi et al [6] proposed a novel Adaptive Contextual Phishing Detection (ACPD) system that combines various techniques of machine learning, deep learning, and user behavior profiling to enhance email filtering systems against phishing attacks. Although ACPD is efficient in a

controlled environment, it lacks scalability and flexibility in validating its results in a dynamic environment.

Zhao et al [7] developed Fewshing, a few-shot learning method for phishing email detection, which showed impressive accuracy and scalability. Though its scalability is impressive, it has not been tested empirically, and like other AI-based systems, its effectiveness is partially restricted, especially in practical situations. Though it performs well in a controlled environment, it does not focus on some of the most important issues like data integrity, trust, and traceability of emails, which are of utmost importance for the comprehensive security of emails.

Vasudevan et al [8] presented a multi-modal approach for spam detection using KNN and SVM techniques with an accuracy of 99.23%. Although it is highly accurate, it is not robust in handling class imbalance and is not comparable with more recent deep learning techniques for its adaptability in modern phishing scenarios.

Heiding et al [9] studied the ways in which attackers use large language models (LLMs) like GPT4 to create authentic phishing emails. This study indicates the rising level of sophistication of phishing attacks and emphasizes the need for developing advanced AI-based solutions to counter these attacks effectively

Pham et al [10] proposed a concept of Adversarial Website Generation using GANs to investigate the robustness of ML models against phishing attacks. Their work highlighted the significant need to protect AI systems from adversarial attacks, which are a threat to AI and must be addressed.

DSouza et al [11] have presented the effectiveness of phishing website detection through comparative analysis of various feature extraction and machine learning techniques in different operational modes: offline, batch, and incremental modes. Although it is more focused on performance parameters like F1 score and time complexity, it offers valuable insights into efficiency considerations. Further analysis is required for more generalized scenarios.

Sahingoz et al [12] developed a phishing detection system based on deep learning technology, which uses convolutional neural networks (CNN) to detect phishing attacks with high accuracy. Although the model has shown good results, the authors did not sufficiently consider the issue of scalability and flexibility in real-world scenarios.

Zhao et al [13] introduced Fewshing, a few-shot learning technique that aims to increase accuracy in detecting phishing emails even when there is a limited amount of training data. Fewshing proved to be efficient in achieving high accuracy and in leveraging contrastive learning mechanisms, although its scalability is yet to be tested.

Jovanovic et al [14] suggested a hybrid phishing detection system, which utilizes novel feature selection algorithms and XGBoost optimization to enhance the adaptability of the system against existing phishing attacks. Although the system has improved adaptability and precision, further scalability and testing need to be done to validate the effectiveness of the system.

Khalid et al [15] have presented a stacking model known as LogiTriBlend, which utilizes various classifiers of machine learning and vectorization techniques to reduce false positives in phishing email identification. Although it is effective in terms of precision, it needs to be further benchmarked and validated to ascertain its robustness.

Asiri et al [16] did a review of smart solutions for detecting HTML phishing attacks with emphasis on the challenges in feature extraction and preprocessing for more advanced techniques in deep learning. From the analysis, it is clear that there are limitations that must be worked on for better and more efficient phishing detection.

He et al [17] developed a feature extension-based phishing URL detection model, combining the TextRank algorithm, CNN, and BERT layers to enhance the detection of phishing websites. Although the effectiveness of the developed model is impressive, further scalability and deployment of the model are required to enhance its robustness.

2.2 CRYPTOGRAPHIC DEFENSES & SPECIALIZED SOLUTIONS

Karim et al [18] improved the process of sender verification during phishing detection through the integration of cryptographic approaches like SPF and DKIM with hybrid models of machine learning, which analyze phishing URLs. This improved the process of email authentication and detection, and the approach must be further tested for scalability.

Kalabarige et al [19] focused on hybrid phishing websites' detection methods that combine feature selection techniques with cryptographic authentication to boost detection rate effectiveness. Although it is more accurate and secure, it needs more testing for its scalability and practical application for its effectiveness against phishing attacks.

Luo et al [20] proposed a phishing account detection method for Ethereum transactions that utilizes cryptographic elements in analyzing the transaction history. This shows how cryptography can contribute to better phishing detection capabilities, especially in environments like blockchain. However, more tests and evaluations must be done to assess its effectiveness against various phishing attacks.

Brezeanu et al [21] have developed Phish Fighter, which is a phishing detection system using structural pattern analysis and clustering methods to provide stronger resilience against zero-day

phishing attacks. Although this system is able to provide stronger resilience against zero-day phishing attacks, more work is required to validate its robustness.

Castano et al [22] have tackled this issue of insufficient real-world phishing kits with their PhiKitA dataset, which is specifically designed to offer a systematic tool to analyze phishing attacks. The authors have stressed the importance of this tool in developing effective detection mechanisms, although more work is required to extend this tool to cover all types of phishing attacks.

2.3 BLOCKCHAIN-BASED APPROACHES

Gupta et al [23] suggested the BERT-TLO hybrid model for phishing detection, which utilizes blockchain data for security enhancement. Although the model has shown promising phishing detection capabilities, the use of blockchain data has been a concern regarding its adaptability in heterogeneous environments.

Rachad et al [24] presented a decentralized and privacy-preserving email system with the aim of boosting security in phishing attacks. Nevertheless, it is not clear if the proposed system has been implemented and performance-tested in various environments.

Al-Karaki et al [25] proposed that blockchain technology could be utilized in re-engineering existing email protocols to prevent phishing attacks. Although this idea illustrates the potential of blockchain technology in improving email security, it is still speculative and not tested in a practical environment

Xia et al [26] have applied ego graph embeddings to enhance phishing detection using blockchain. Although this approach has shown promise in improving the accuracy of phishing detection, it has not shown promise in terms of scalability and real-time processing.

Ray et al [27] employed quantum machine learning (QML) for phishing detection using the Ethereum platform and QRAC encoding. Though this method has been innovative in the use of quantum computing for blockchain, there has been no testing of the system against adversarial examples, thereby its practicality has not been proven.

Hossain et al [28] employed smart contracts and elliptic curve cryptography to facilitate a secure key exchange between senders and receivers using PGP, thereby ensuring phishing-resistant communication. Although this enhances cryptographic security, the study does not present scalability analysis or benchmarking results, thus its practicality remains questionable.

Pandeewari et al [29] proposed encrypting email contents and metadata and recording it in a blockchain ledger for enhanced phishing resistance. Although the approach demonstrates the potential of blockchain in secure communication, it was not evaluated for its efficiency in real-time and lacked essential implementation aspects.

Xu et al [30] developed a system called EWDPS, a hybrid system that integrates machine learning with the analysis of smart contract behavior to identify phishing attacks on Ethereum. Although it offers an early warning system through its profiling of transaction patterns, it still has limitations with regard to deployment validation, has limited generalization of scams, and lacks adversarial robustness.

Kang and Buu [31] used graph anomaly detection techniques in blockchain platforms to solve the phishing issue in Bitcoin transactions. This approach proved the efficacy of anomaly-based graph analysis in detecting malicious activity; however, more tests for its scalability and robustness against attacks must be conducted to assess its applicability in various environments.

CHAPTER 3

PRESENT WORK

3.1 PROBLEM FORMULATION

One of the most prevalent cyber security threats to individuals and enterprises all over the world today is a phishing attack. This is because the attacker uses email as a medium of deceit in order to steal sensitive personal information from the victim, such as financial data and passwords. The technique used to conduct phishing has evolved significantly owing to the rapid evolution of communication channels on the Internet, thereby complicating the entire process and making it harder to be detected. Traditional methods of filtering email and applying detection rules have become ineffective against new techniques of phishing involving malware and information engineering.

However, the traditional phishing detection tool relies on the use of signature-based filtering tools, heuristic-based detections, and blacklists. Even though the traditional detection technique is efficient in detecting malicious patterns that are already known, they do not work well in detecting newly created phishing threats. Attackers have been altering their domains, email format, and URL to avoid being detected by the traditional technique. Consequently, traditional detection techniques cannot efficiently detect phishing threats due to changes made by attackers.

To address such limitations, several methods based on machine learning techniques have been developed to identify phishing attacks. Methods based on machine learning algorithms utilize various features of emails like headers, content, links, and attachments to determine whether an email is harmful or harmless. Machine learning algorithms are highly accurate and adaptive enough to detect new phishing attacks. Although there are many benefits offered by AI-based detection systems, several limitations exist in these types of detection systems. The main challenges faced by machine learning-based detection systems include false positives, adversarial attacks, and centralization of detection services. Machine learning-based detection systems cannot verify the legitimacy of email communications.

Cryptographic techniques such as SPF, DKIM, and DMARC can normally be used for authentication of the sender's credentials and to thwart any attacks involving domain spoofing. While these cryptographic techniques improve the security of e-mail communications, they are incapable of identifying the presence of phishing content in the body of an e-mail. For example, when the phishing e-mail originates from either a genuine or compromised domain, the above cryptographic techniques are unable to detect a phishing attack.

Moreover, the creation of new cyber security methods, which utilize the concept of block chain technology, provides security as well as data storage, yet these methods do not have any smart system that could help identify phishing attacks in real-time. Consequently, the current methods provide either threat detection or communication security, yet never both together.

Therefore, it is essential that there be developed a model which incorporates intelligent detection methods such as those using machine learning as well as security mechanisms which are tamper-resistant in nature through the use of blockchain technology. Such a system would not only increase the effectiveness of phishing detection, but also instill confidence in email communication systems. This paper seeks to fill this research gap by designing a model which incorporates machine learning and blockchain technology in phishing detection.

3.2 OBJECTIVES OF THE STUDY

The primary objective of this study is to develop an intelligent phishing detection system based on email communication by integrating machine learning and blockchain technologies. It should be acknowledged that phishing threats can be dynamic, and the detection mechanisms might not be capable of achieving accurate phishing detection and verifying the security of communications. As such, there is a need to develop a hybrid framework to enhance phishing detection accuracy and ensure safe communication.

The specific objectives of this research are as follows:

1. To design and implement a phishing detection model based on machine learning.

This objective is associated with the creation of a classifier model that could be applied to identify phishing emails through email characteristics like email header, URL, body, and sender details. The classifier uses different machine learning models to categorize emails into three groups: phishing, spam, and legitimate emails.

2. To extract and analyse relevant features from email data to ensure efficient phishing detection.

The detection process of phishing involves the analysis of features present in the email messages that are considered suspicious. These features can be extracted from different parts of an email message, which includes domain, hyperlinks, email headers, and features based on the content.

3. To integrate blockchain technology for secure email verification and communication integrity.

The second goal of the research is to integrate the concept of blockchain technology into the phishing detection model. Blockchain refers to a decentralized platform that can be utilized for

secure storage of emails based on their metadata. It can be achieved through hashing of the email content and subsequent storage in the blockchain system.

4. To develop a hybrid AI-Blockchain framework for enhanced phishing detection.

The purpose of the study is to design a hybrid cybersecurity system that integrates artificial intelligence and blockchain technologies in email security and detection of cyber threats. In this regard, artificial intelligence will be used for email classification, whereas blockchain technology will be deployed for secure record-keeping of verified emails, preventing any form of manipulation.

5. To assess the performance of the proposed system using the appropriate evaluation metrics.

For the evaluation of the performance of the suggested hybrid model, specific parameters are considered such as accuracy, precision, recall, and F1-score. These parameters are useful for determining the performance of the suggested model so that the system is capable of identifying phishing emails without making any error.

6. To analyse the effectiveness and reliability of the hybrid detection framework.

This is the last objective, and the goal here would be to examine how effective this new approach in terms of detecting phishings would perform when contrasted with conventional approaches. This will go far in ensuring that the proposed solution can actually be applied in current environments.

3.3 RESEARCH METHODOLOGY

This research has employed a hybrid model in cybersecurity, which uses machine learning techniques in combination with secure communication to detect phishing emails. The methodology employed in this research has been developed in two phases. The first phase has focused on the conceptual framework, which uses artificial intelligence in combination with cryptographic authentication to detect phishing emails. The second phase has built upon this research by implementing an enhanced version of the framework, which uses artificial intelligence in combination with blockchain technology.

3.3.1 Initial Hybrid Detection Approach (AI and Cryptography)

In the initial phase of this study, emphasis was placed on the examination of a mixed detection system which would make use of both machine learning and cryptography in order to authenticate email senders. The aim was to improve the detection rate of the phishing attacks by integrating intelligent content analysis along with sender verification.

In this approach, machine learning algorithms have been suggested to analyze various attributes of the emails, such as header, URL, attachments, and content. Such attributes are believed to be vital for discovering patterns that characterize the phishing attacks, as they can generally be regarded as suspicious. Phishing or legitimate classification of emails was performed using a random forest, which is a machine learning algorithm.

Apart from applying machine learning in data analysis, cryptography techniques were also employed to authenticate the identity of the sender and to safeguard the integrity of the emails. Security techniques applied to prevent emails from being spoofed include the Sender Policy Framework (SPF), DomainKeys Identified Mail (DKIM), and Domain-based Message Authentication, Reporting, and Conformance (DMARC). This is done to confirm that emails originate from legitimate sources and are not altered in any manner during transmission.

Nonetheless, there was a theoretical basis provided by this AI-based cryptographical approach to enhance phishing detection. However, this approach was only theoretical because it had not yet been put into practice. In addition, the cryptographic verification methods cannot detect phishing attacks that come from genuine domains, which have been compromised. Thus, there was an urgent need to develop a better approach that would make phishing detection easier and also allow for intelligent threat detection.

3.3.2 PROPOSED HYBRID FRAMEWORK (AI AND BLOCKCHAIN)

In comparison with the first methodology proposed above, in this research paper a more advanced method, which incorporates artificial intelligence and the blockchain technology, is offered for the detection of phishing attacks and validation of emails.

For this purpose, a two-layer model has been created, including an intelligent detection layer and a secure verification layer. The former will utilize machine learning algorithms in order to classify emails, and the latter will make use of blockchain technology.

In regard to the process of detection, the incoming e-mails are analyzed through some features like e-mail headers, content, links, attachments, and also the meta-data. There will be some data preprocessing and extraction of features for getting the required information in structured format. Some machine learning techniques are used to make the classification process effective. The examples of such machine learning techniques are: Logistic regression, Random Forest, SVM (Support vector machine), and LSTM (Long short-term memory).

Regarding the verification stage, blockchain is utilized to secure integrity of the verified emails. Following the successful email detection stage, a hash is generated based on email metadata through SHA-256 algorithm. After that, this hash is added to the blockchain with the help of smart contracts.

The blockchain layer brings a number of advantages, such as decentralization, eliminating centralization; immutability, ensuring there are no unauthorized modifications; and traceability allowing tracing communications. Another advantage provided by the blockchain layer is storage of data as hashes.

Combining machine learning and blockchain creates a robust system that can detect phishing and guarantee secure communication.

3.3.3 System Workflow

The workflow of the suggested system illustrates the relationship between the two systems.

The first process involves email acquisition, which entails the email header, email body, URL, and email attachment. Once the data acquisition process is completed, pre-processing follows, whereby noise elimination and normalization occur.

When the pre-processing process is completed, the process of feature extraction takes place. After the completion of the process of feature extraction, these features will be passed to the machine learning classifier. The machine learning classifier will classify the email messages into three groups: phishing, spam, and normal email messages.

After classification, a decision rule is formulated based on the outcome of the classification process. The email classified as phishing may be blocked or flagged. On the contrary, the email classified as a legitimate one progresses to the verification phase by blockchain technology, where the email's metadata is hashed through the SHA-256 algorithm and saved via smart contracts.

The last phase involves generating the output from the classification process and confidence level of the prediction generated through machine learning algorithms.

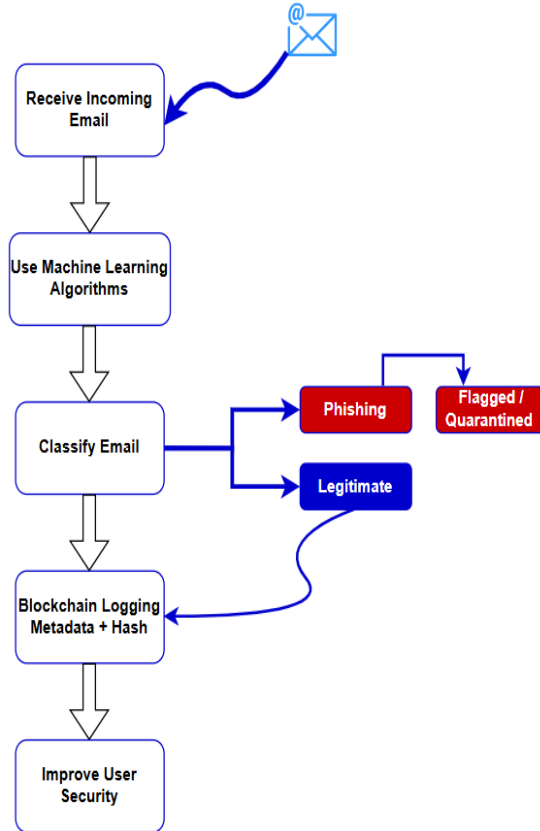


Figure 1: System Integration and Architecture Flowchart

3.3.4 Dataset Collection

In this study, the dataset that will be used is labeled samples of emails categorized as phishing, spam, and legitimate emails. The dataset is acquired through reliable resources that are commonly known for phishing data.

The main datasets used in this study include the SpamAssassin Public Corpus and the phishing datasets found in Kaggle and other online public databases. These datasets consist of different email samples ranging from the benign ones to malicious emails.

Some of the aspects considered in the dataset include email header, body, URL, attachment, and metadata, among others. These make it easy for the system to learn better. There is also separation of training and test data sets in the data set.

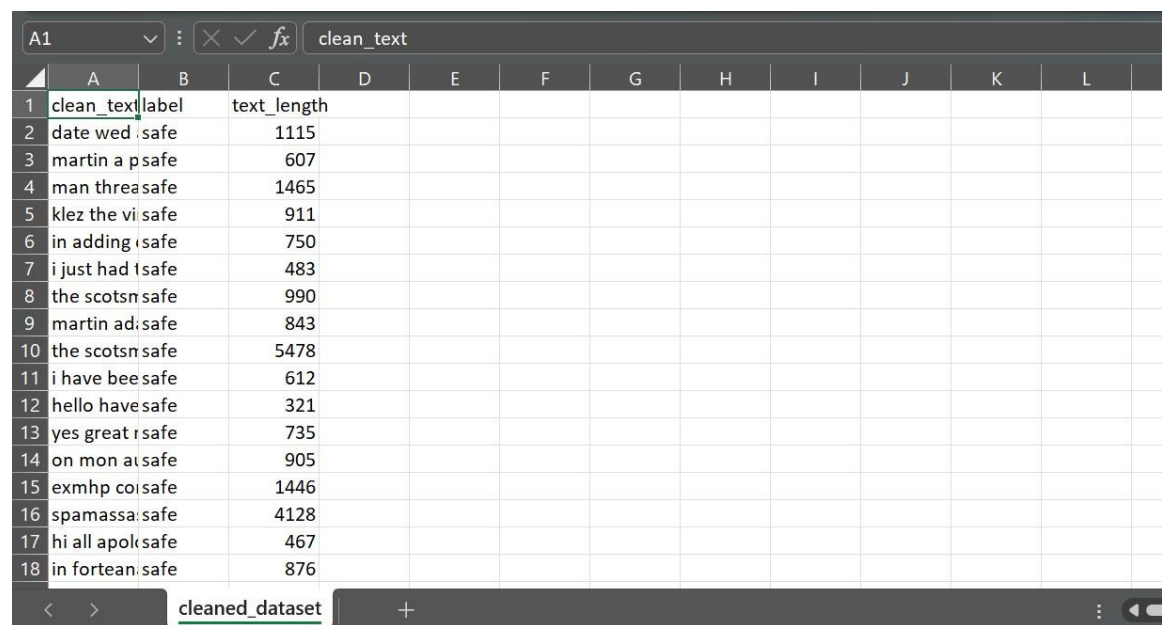
3.3.5 Data Preprocessing

Data preprocessing is an important task when transforming raw data into structured data suitable for machine learning applications.

The steps involved in data preprocessing include the removal of duplicated and unwanted data, preprocessing of emails through removal of HTML tags, special characters, and noises from the

emails, as well as converting the texts to lower case. Text tokenization is used to split the text into individual tokens before stop words are removed. Finally, lemmatization reduces words to their base form.

Lastly, text is converted into numbers using TF-IDF vectorization.



	A	B	C	D	E	F	G	H	I	J	K	L
1	clean_text	label	text_length									
2	date wed	safe	1115									
3	martin a p	safe	607									
4	man threa	safe	1465									
5	klez the vi	safe	911									
6	in adding	safe	750									
7	i just had	safe	483									
8	the scotsn	safe	990									
9	martin ad	safe	843									
10	the scotsn	safe	5478									
11	i have bee	safe	612									
12	hello have	safe	321									
13	yes great	safe	735									
14	on mon a	safe	905									
15	exmhp co	safe	1446									
16	spamassa	safe	4128									
17	hi all ap	safe	467									
18	in fortea	safe	876									

Figure 2: Text Conversion using TF-IDF vectorization

3.3.6 Feature Extraction

Features are extracted to recognize useful attributes that can differentiate between a phishing email and a normal one.

The following features have been extracted: header features, which consist of the domain of the sender and IP address; content features, which include suspicious keywords and urgency; URL features, which contain domain age and URL length; and attachment features, which include the type of file used and embedded scripts.

Such features offer insightful information for classification purposes.

3.3.7 Machine Learning Model

Several machine learning algorithms are used for email classification. They are Logistics Regression, Random Forest, SVM (Support Vector Machine), and LSTM (Long Short-Term Memory).

All of the mentioned models are trained using data with labels, and the quality of training is measured by such metrics as accuracy, precision, recall, and F1-score. All of these metrics give an extensive evaluation of the model's performance, especially for an unbalanced dataset.

The chosen algorithm should be able to detect emails correctly without errors.

3.3.8 Blockchain Integration

The technology of Blockchain is utilized to ensure the secure and immutable storage of the validated emails on the platform.

If the email is determined as valid, the SHA-256 hash for the metadata associated with the email will be generated and stored within the Blockchain through the use of smart contracts. The Blockchain will be developed based on the Ethereum network, with Ganache used as a local development environment. Web3.py will be used for interfacing between the platform and the Blockchain.

3.3.9 Web Interface

Web interface is created utilizing Streamlit to ensure user interaction with the system. This interface ensures that the users can enter their email data and see how well the emails are classified as authentic via the help of blockchain technology and even see the confidence score.

The purpose of such an interface is to make sure that there will be no difficulty in interacting with the system.

3.3.10 Tools and Technologies

The implementation of the proposed system utilizes various tools and technologies, as summarized in table below

Table I: Tools and Technology Used

Component	Technology	Purpose
Programming Language	Python 3	Core system development
Machine Learning	Scikit-learn	ML model implementation
Deep Learning	TensorFlow / Keras	LSTM model
Data Processing	Pandas, NumPy	Data manipulation
NLP	NLTK	Text preprocessing
Visualization	Matplotlib, Seaborn	Performance analysis
Blockchain	Ethereum	Decentralized storage
Blockchain Network	Ganache	Local testing
Smart Contracts	Solidity	Blockchain logic
Blockchain Interface	Web3.py	Communication
Web Interface	Streamlit	User interaction

Development	Jupyter Notebook	Model development
-------------	------------------	-------------------

CHAPTER 4

RESULTS AND DISCUSSION

This chapter will discuss the experimental results for the suggested hybrid phishing detection system. The purpose of this chapter is to test the suggested hybrid system and show its effectiveness in detecting phishing emails and secure communication.

The evaluation procedure has three main factors that will be discussed. The first factor will include the discussion about the efficiency of the machine learning algorithms, which will include accuracy, precision, recall, and F1-score.

Second, we will evaluate the performance of the blockchain component of the system, taking into account its effectiveness in storing emails in an immutable way. It is necessary to analyze the operation of smart contracts and the hash function that can be used to store emails.

Third, we should test the developed web application, which shows how users can work with our system and detect phishing emails using blockchain technologies.

In conclusion, we believe it is appropriate to pay attention to the following problems.

4.1 EXPERIMENTAL SETUP

The experimental design is intended to test the hybrid phishing detection system that combines machine learning and blockchain technologies.

Main features of the experimental design include:

- **Dataset:**

The dataset comprises phishing, spam, and legitimate emails with their labels. It contains headers, bodies, URLs, attachments, and other metadata.

	A	B	C	D	E	F	G	H	I
	sender	receiver	date	subject	body	urls	label		
1	MR. JAME	webmaste	Thu, 31 Oc	URGENT B	FROM:M	0	1		
2	Mr. Ben Si	R@M	Thu, 31 Oc	URGENT A	Dear	0	1		
3	PRINCE O	webmaste	Thu, 31 Oc	GOOD DA	FROM	0	1		
4	PRINCE O	webmaste	Thu, 31 Oc	GOOD DA	FROM	0	1		
5	Maryam A	R@M	Fri, 01 Nov	I Need Yo	Dear sir,	0	1		
6	Kuta Davi	davidkuta	Sat, 02 No	Partnershi	ATTENTI	0	1		
7	Barrister tunde dosumu	<tunde	Urgent Att	Dear Sir,		1	1		
8	William D	webmaste	Sun, 03 Nc	URGENT B	FROM:	0	1		
9	MR USMA R	@M	Mon, 04 N	THANK YO	CHALLENG	0	1		
10	Tunde Dosumu	<barrister_td@	Urgent As:	Dear Sir,		1	1		
11	MR TEMI J	R@E	Tue, 06 Nc	IMPORTAI	FROM:	0	1		
12	Dr.Sam jo	R@M	Fri, 08 Nov	URGENT A	FROM	0	1		
13	p_brown2	webmaste	Fri, 08 Nov	From: Bar		0	1		
14	mic_k1@	f R@M	Mon, 11 N	MICHAEL	From;Mr.	0	1		
15	COL. MICH	R@E	Tue, 13 Nc	*****SPA	FROM:	0	1		
16	MRS MAR	webmaste	Thu, 14 Nc	TRUST TR	/Dear Sir,	0	1		
17	DR. ANAY	webmaste	Fri, 15 Nov	REQUEST	/FROM:	0	1		

Figure 3: Content of the Nigerian_Fraud Emails Dataset

	A	B	C	D	E	F	G	H	
	sender	receiver	date	subject	body	label	urls		
1	Robert Elz	Chris Garr	Thu, 22 Au	Re: New S	Date:	0	1		
2	Steve Burt	"zzzzteana	Thu, 22 Au	[zzzzteana	Martin A	0	1		
3	"Tim Chap	zzzzteana	Thu, 22 Au	[zzzzteana	Man	0	1		
4	Monty Sol	undisclose	Thu, 22 Au	[IRR] Klez: Klez: The		0	1		
5	Stewart Sr	zzzzteana	(Thu, 22 Au	Re: [zzzzte	> in	0	1		
6	"Martin A	zzzzteana	(Thu, 22 Au	Re: [zzzzte	> I just	0	1		
7	"Martin A	zzzzteana	(Thu, 22 Au	[zzzzteana	The	0	1		
8	Stewart Sr	zzzzteana	(Thu, 22 Au	Re: [zzzzte	Martin Ad	0	1		
9	"Martin A	zzzzteana	(Thu, 22 Au	[zzzzteana	The	0	1		
10	"NOI Adm	<spamass	.Thu, 22 Au	[SATalk]	S/I have	0	1		
11	David H=?	<spamass	.Thu, 22 Au	[SAdev]	In Hello,	0	1		
12	Marc Perk	Theo Van	Thu, 22 Au	Re: [SAdev	Yes -	0	1		
13	"John P. L	linux-raid	(Thu, 22 Au	[ILUG] Re: On Mon,		0	1		
14	Chris Garr	Robert Elz	Thu, 22 Au	Re: New S	---=_Exmt	0	1		
15	Owen Byr	fork@spai	Thu, 22 Au	The case f	SpamAss	0	1		
16	Bernard Miu	<iuu@t	Thu, 22 Au	[IIU] Eirco	Hi all,	0	1		
17	"uncle_sl	zzzzteana	(Thu, 22 Au	[zzzzteana	--- In forte	0	1		

Figure 4: Content of the SpamAssasin Email Dataset

- **Data Split:**

Training and testing datasets follow the 80:20 proportion.

- **Data Preprocessing:**

This step involves data cleaning, tokenization, removal of stop words, lemmatization, and feature extraction by applying TF-IDF.

	A	B	C	D	E	F	G	H	I	J	K	L
1	clean_text	label	text_length									
2	date wed	safe	1115									
3	martin a p	safe	607									
4	man threa	safe	1465									
5	klez the v	safe	911									
6	in adding	safe	750									
7	i just had	safe	483									
8	the scotsn	safe	990									
9	martin ad	safe	843									
10	the scotsn	safe	5478									
11	i have bee	safe	612									
12	hello have	safe	321									
13	yes great	safe	735									
14	on mon a	safe	905									
15	exmhp co	safe	1446									
16	spamassa	safe	4128									
17	hi all ap	safe	467									
18	in fortea	safe	876									

Figure 5: Cleaned Dataset

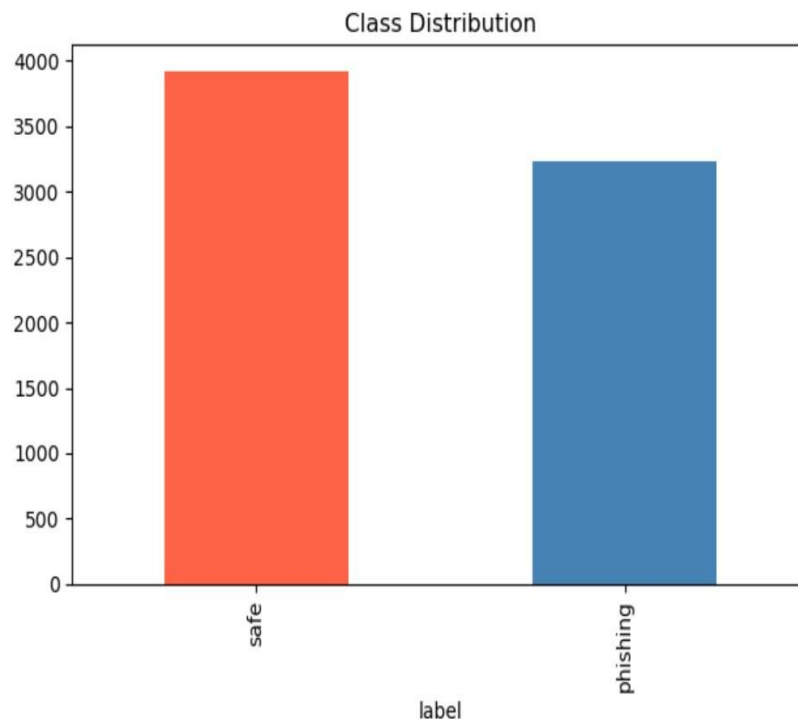


Figure 6: Class Distribution for Safe and Phishing Emails

- **Machine Learning Models:**

Different models are applied: Logistic Regression, Random Forest, SVM, and LSTM.

- **Classification Approach:**

The task is approached with the use of two kinds of classification:

Binary classification (phishing and legitimate)

Multiclass classification (phishing, spam, and legitimate)

- **Evaluation Metrics:**

Model performance is assessed on the basis of accuracy, precision, recall, and F1score.

- **Blockchain Technology:**

Ethereum (Ganache) was chosen as the blockchain environment; the smart contract is developed in Solidity; interactions are realized with the help of Web3.py.

- **System Implementation:**

Integration of the machine learning model and blockchain module into one system takes place.

- **User Interface:**

The web interface is created using Streamlit.

4.2 PERFORMANCE EVALUATION METRICS

The performance of the machine learning models is measured based on the metrics used in classification problems. The metrics give a clear insight into how effective the proposed system is.

- **Accuracy:** indicates the accuracy of the classifier
- **Precision:** indicates the number of phishing emails identified as phishing out of all the phishing emails identified
- **Recall:** indicates the capability of the classifier to detect phishing emails correctly
- **F1-score:** indicates the precision and recall together

These metrics help in choosing the best model for detecting phishing.

4.3 RESULTS OF MACHINE LEARNING MODELS

The results of the training of the various machine learning models which aided in the implementation of the research paper can be found in the figures below.

4.3.1 LOGISTIC REGRESSION

Logistic Regression				
	precision	recall	f1-score	support
phishing	0.99	0.97	0.98	646
safe	0.98	0.99	0.98	785
accuracy			0.98	1431
macro avg	0.98	0.98	0.98	1431
weighted avg	0.98	0.98	0.98	1431

Figure 7: The Performance Evaluation Metrics for Logistic Regression

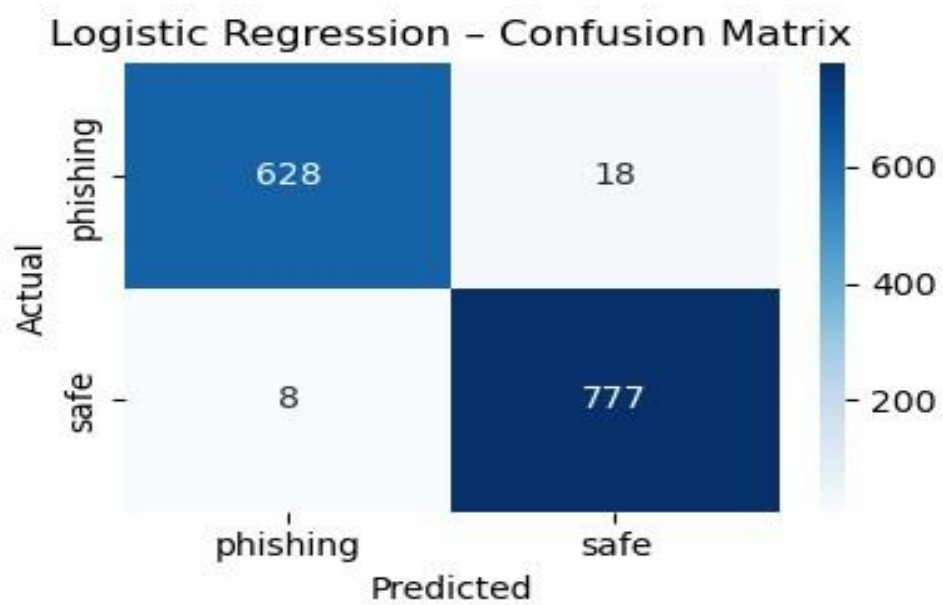


Figure 8: Confusion Matrix for Logistic Regression

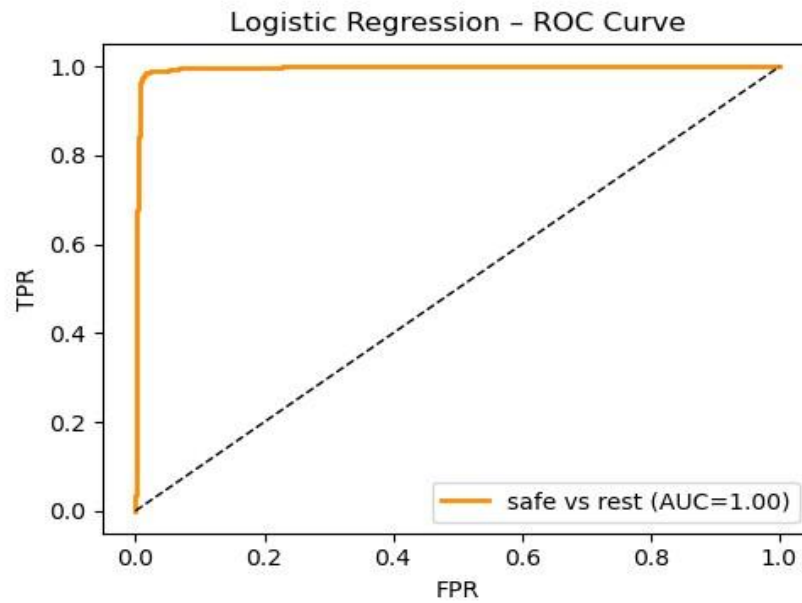


Figure 9: ROC Curve for Logistic Regression

4.3.2 RANDOM FOREST

Random Forest				
	precision	recall	f1-score	support
phishing	0.97	0.99	0.98	646
safe	0.99	0.98	0.98	785
accuracy			0.98	1431
macro avg	0.98	0.98	0.98	1431
weighted avg	0.98	0.98	0.98	1431

Figure 10: The Performance Evaluation Metrics for Random Forest

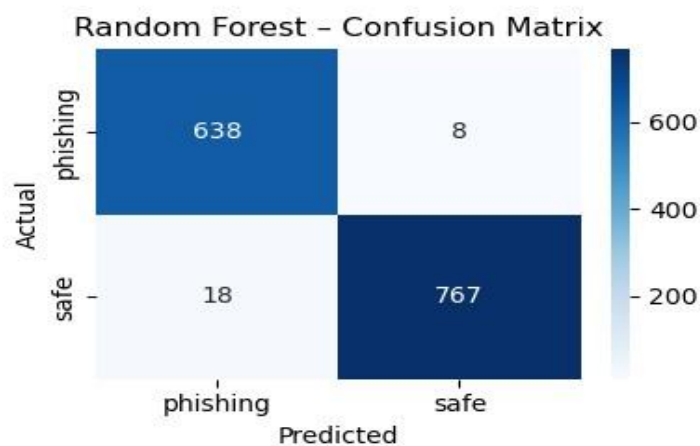


Figure 11: Confusion Matrix for Random Forest

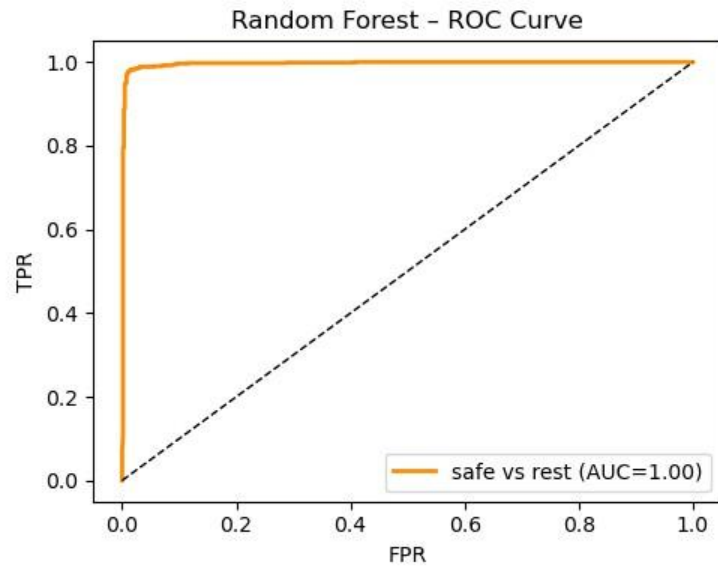


Figure 12: Roc Curve for Random Forest

4.3.3 SVM

SVM				
	precision	recall	f1-score	support
phishing	0.99	0.98	0.99	646
safe	0.98	0.99	0.99	785
accuracy			0.99	1431
macro avg	0.99	0.99	0.99	1431
weighted avg	0.99	0.99	0.99	1431

Figure 13: The Performance Evaluation Metrics for SVM

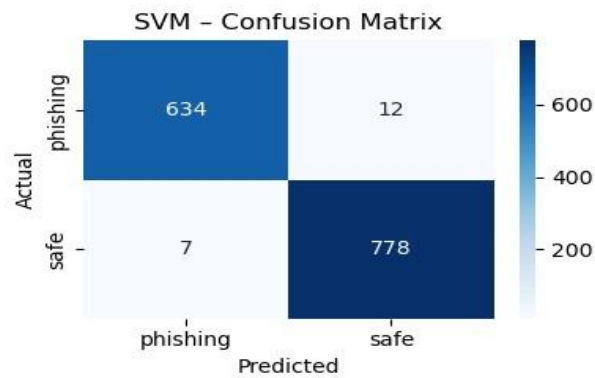


Figure 14: Confusion Matrix for SVM

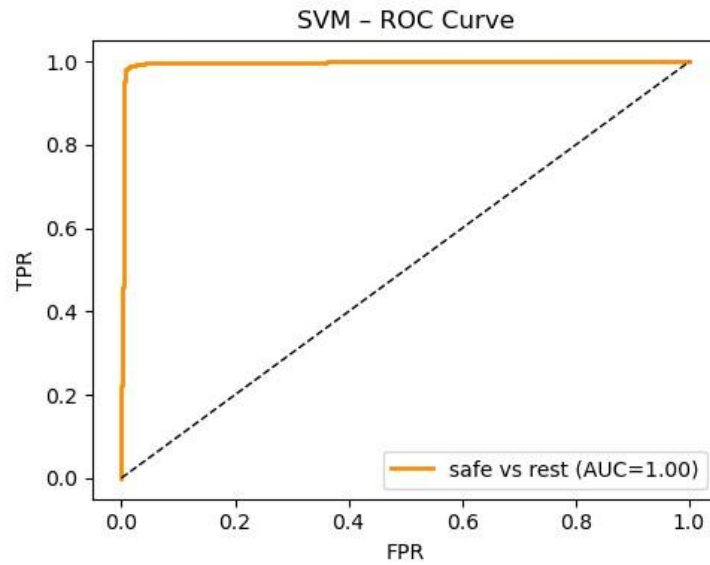


Figure 15: ROC Curve for SVM

4.3.4 LSTM

LSTM	precision	recall	f1-score	support
phishing	0.95	0.89	0.92	646
safe	0.91	0.96	0.94	785
accuracy			0.93	1431
macro avg	0.93	0.93	0.93	1431
weighted avg	0.93	0.93	0.93	1431

Figure 16: The Performance Evaluation Metrics for LSTM

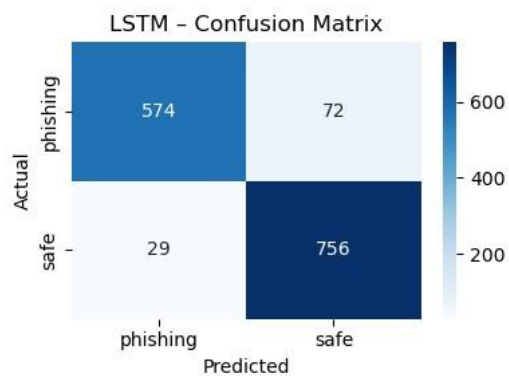


Figure 17: Confusion Matrix for LSTM

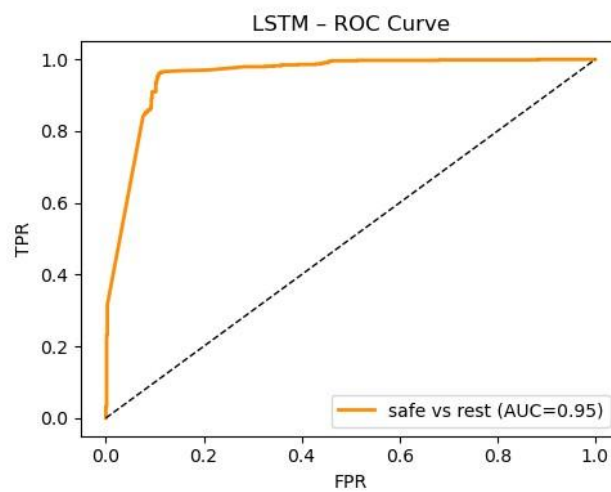


Figure 18: ROC Curve for LSTM

4.3.5 MODEL PERFORMANCE COMPARISON

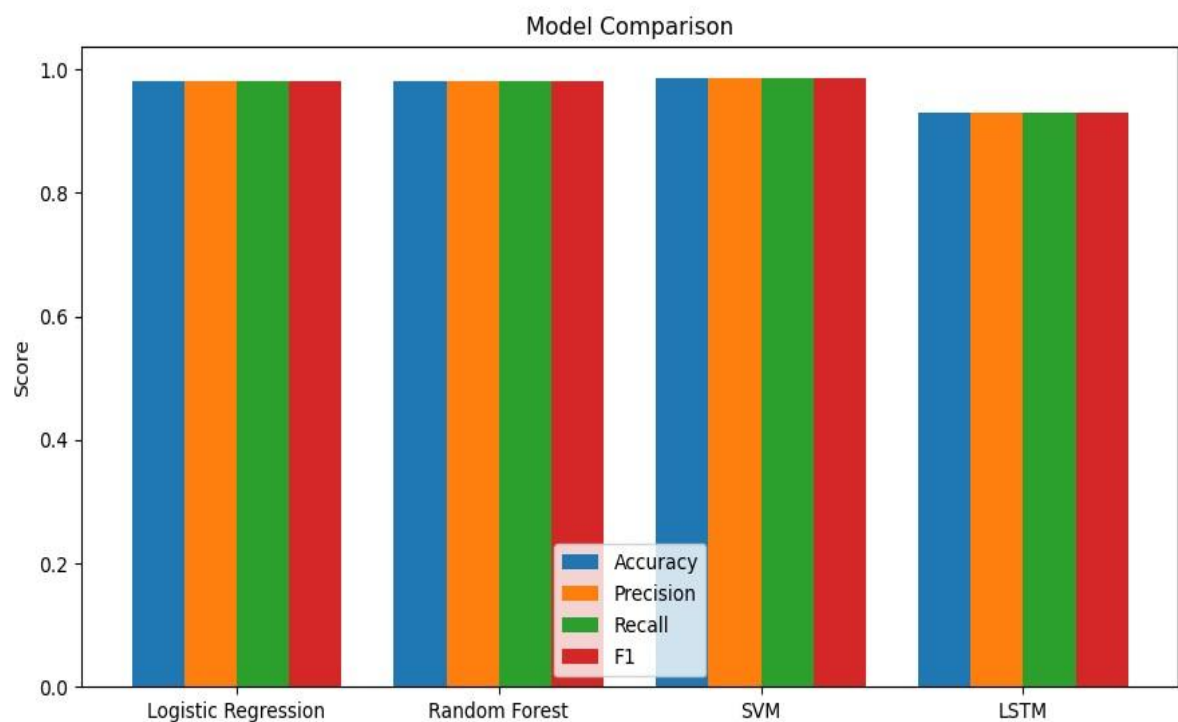


Figure 19: Bar Graph Showing the various ML models Performance Comparison

4.3.6 MACHINE LEARNING PERFORMANCE COMPARISON

Table II: Machine Learning Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	98%	0.98	0.98	0.98
Random Forest	98%	0.98	0.98	0.98
SVM	99%	0.99	0.99	0.99
LSTM	93%	0.93	0.93	0.93

As can be observed from the results shown in Table II above, SVM gives the best performance among all the other models. On the other hand, Logistic Regression and Random Forest perform strongly and have comparable performance as well.

LSTM has the lowest performance when compared to other models. This is attributed to the type of data set and the way features have been represented in this case.

It can be observed that the performance of traditional machine learning techniques outperforms the one of deep learning techniques.

4.4 DISCUSSION OF RESULTS

It can be seen from Table II that the SVM classifier performs better than the other two classifiers with regard to their accuracy, precision, recall, and F1 measure scores. Therefore, it can be concluded that the SVM classifier performs excellently in differentiating between phishing emails and legitimate emails.

It can be seen that both Logistic Regression and Random Forest provide good accuracy. The similarity in the performance of these two models indicates their effectiveness in performing phishing detection. Moreover, the two models are capable of learning effectively from structured features available in emails.

In contrast, the performance of the LSTM is quite poor in comparison to the other models. This may be attributed to the fact that deep learning algorithms require huge amounts of data and a more sophisticated way of representing features in order to achieve high-level performance. In this case, the classic learning algorithms appear to have been made specifically for such a dataset.

From the results obtained, it can be seen that classical machine learning models perform better than deep learning models. This may be attributed to the structure of the dataset and the ability of classical machine learning models to exploit features effectively.

It can therefore be concluded that classical machine learning models perform much better than deep learning models in phishing detection systems.

4.5 BLOCKCHAIN RESULTS

This blockchain-based approach in the proposed design allows ensuring that email data verification and storage processes are secure, transparent, and not vulnerable to any malicious actions.

Within this approach, after an email being declared legitimate by the machine learning model, a hash of metadata from such emails is created using the SHA-256 hashing function depicted in figure 20. After that, this metadata hash can be stored in the blockchain via smart contracts in the Ethereum environment.

This blockchain-based approach is realized by means of Ganache software used as the local blockchain environment, as well as Solidity and Web3.py for developing smart contracts and interacting with the blockchain, respectively.

The findings show that the blockchain has the capability to securely store emails. Every transaction is entered into the block, and the confirmation of every entry ensures its immutability. The ability to verify the authenticity of the transactions through hashing shows that email transactions can be easily verified. Figure 21 and 22 shows email transactions that has been hashed.

It takes low latency to perform the operations on the blockchain network, hence making it efficient in near real-time operations. Moreover, the fact that hashes and not the content are stored ensures data privacy and security.

Finally, the blockchain findings confirm that the developed system has provided a good way of securing emails and verifying their authenticity alongside machine learning-based phishing detection.

norm.eml
 From : Georgianna Narpati <vbvnzcxmnzxcvxcz798@gmail.com>
 Subject : DHYVN Purchase report on 22/07/2025 OWQI 36589
 Result : SAFE
 Hash stored on blockchain | TxHash: f63e263b7bd71957b09d0144b52c7f49b41c9785076951a2db536cfeebb831d9

norma.eml
 From : NPTEL Newsletter <newsletter-announce+noreply@nptel.iitm.ac.in>
 Subject : You've successfully unsubscribed from newsletter-announce.
 Result : SAFE
 Hash stored on blockchain | TxHash: f22ec569c603e44fc88ca5025c438cb6074e5b7690691f173b6e8c3fd0b78725

phishing_email1.eml
 From : Eyitayo from Utiva <advisor@utiva.co>
 Subject : Can You Commit 3 Days To This?
 Result : SAFE
 Hash stored on blockchain | TxHash: 873b8e08b81bd14ef56c84251c843c9f9b23e2f94ddc7297f3e35eca5061b3a2

phishing_email2.eml
 From : Omowunmi from Utiva <advisor@utiva.co>
 Subject : We Are Live Now - See How to Earn \$100/hr!
 Result : PHISHING
 Phishing email - NOT stored on blockchain.

phishing_email3.eml
 From : Rajayi Raji <rajiajay069@gmail.com>
 Subject : Do you need a job?
 Result : SAFE

Figure 20: Email Classification and Storage on Blockchain

Ganache									
ACCOUNTS					BLOCKS				
TRANSACTIONS					CONTRACTS				
EVENTS					LOGS				
CURRENT BLOCK 13					GAS PRICE 20000000000				
GAS LIMIT 6721975					HARDFORK MERGE				
NETWORK ID 5777					RPC SERVER HTTP://127.0.0.1:7545				
MINING STATUS AUTOMINING					WORKSPACE AI BLOCKCHAIN				
SWITCH									
MNEMONIC ?					HD PATH				
tumble enemy rail illegal inner cook spare general tackle huge icon choose					m44'60'0'0account_index				
ADDRESS	BALANCE	TX COUNT	INDEX						
0xd924721047859C3B53E8A2e6Cfeffe7Df0B6fcd0	99.90 ETH	13	0						
ADDRESS	BALANCE	TX COUNT	INDEX						
0x9EF88368bdE13c993f98454D92e189b867a3C35C	100.00 ETH	0	1						
ADDRESS	BALANCE	TX COUNT	INDEX						
0x98B101B36789E15f432d2Aa40268995100629177	100.00 ETH	0	2						
ADDRESS	BALANCE	TX COUNT	INDEX						
0x6C67D15Ef2255AB6fB47111a35CFF86dD1F41D90	100.00 ETH	0	3						
ADDRESS	BALANCE	TX COUNT	INDEX						
0x965c3C4Ffb0F48f5073E9b8823fB9e75AFc277bd	100.00 ETH	0	4						
ADDRESS	BALANCE	TX COUNT	INDEX						

Figure 21: Ganache Application Platform

ACCOUNTS	BLOCKS	TRANSACTIONS	CONTRACTS	EVENTS	LOGS	SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 13	GAS PRICE 20000000000	GAS LIMIT 6721975	HARDFORK MERGE	NETWORK ID 5777	RPC SERVER HTTP://127.0.0.1:7545	MINING STATUS AUTOMINING
WORKSPACE AI BLOCKCHAIN						
SWITCH ⚙️						
BLOCK 13	MINED ON 2026-04-22 22:29:16				GAS USED 305358	1 TRANSACTION
BLOCK 12	MINED ON 2026-04-22 21:16:13				GAS USED 629012	1 TRANSACTION
BLOCK 11	MINED ON 2026-04-14 20:42:11				GAS USED 299566	1 TRANSACTION
BLOCK 10	MINED ON 2026-04-14 18:40:24				GAS USED 629012	1 TRANSACTION
BLOCK 9	MINED ON 2026-04-14 18:32:45				GAS USED 629012	1 TRANSACTION
BLOCK 8	MINED ON 2026-04-14 16:26:14				GAS USED 283383	1 TRANSACTION
BLOCK 7	MINED ON 2026-04-14 16:26:14				GAS USED 237995	1 TRANSACTION
BLOCK 6	MINED ON 2026-04-14 16:26:14				GAS USED 305358	1 TRANSACTION

Figure 22: Ganache Interface Showing Blockchain Blocks and Transactions Generated by the Proposed System

The figures above illustrate the Ganache interface for the Ethereum blockchain. The blocks contain details of transactions such as block numbers, timestamps, and amount of gas consumed during the mining process.

These transactions correspond to the storage of hashed information on emails that have been validated by the system. The blocks provide immutable storage for the records of email transactions stored in the blockchain.

The consumption of gas by the blocks indicates that the blockchain component is working.

Therefore, the figures above shows that the system manages to store records of validated emails on the blockchain.

4.6 DISCUSSION OF BLOCKCHAIN RESULTS

The findings obtained using the blockchain illustrate the effectiveness of incorporating blockchain technology in enhancing the security and reliability of the phishing detection system proposed.

First, one of the main advantages that have been experienced with the use of blockchain technology is the inability to alter the information that can be stored in the blockchain. This means that with the use of hashed emails being uploaded into the blockchain system, any attempt to change the information will not be successful.

Secondly, all the emails available in the blockchain can be tracked. The use of SHA-256 hashing ensures that all the emails' data are safe since actual emails will not be uploaded to the blockchain. Instead, the hash values of the emails will be used. This provides an effective combination of security and confidentiality.

Furthermore, since the blockchain technology is decentralized, there is no need for a centralized system in the process, which would result in risks.

On the other hand, some disadvantages are also there. The use of blockchain could result in increased computations and slower speed in comparison to the absence of blockchain. However, it must be kept in mind that the adoption of local blockchain proves that the system can work close to real time.

It can thus be concluded that the use of blockchain technology ensures adequate security of the confidentiality of emails and ensures an effective means of communication. Furthermore, the use of blockchain and machine learning will ensure a dependable system to detect phishing.

4.7 WEB INTERFACE RESULTS

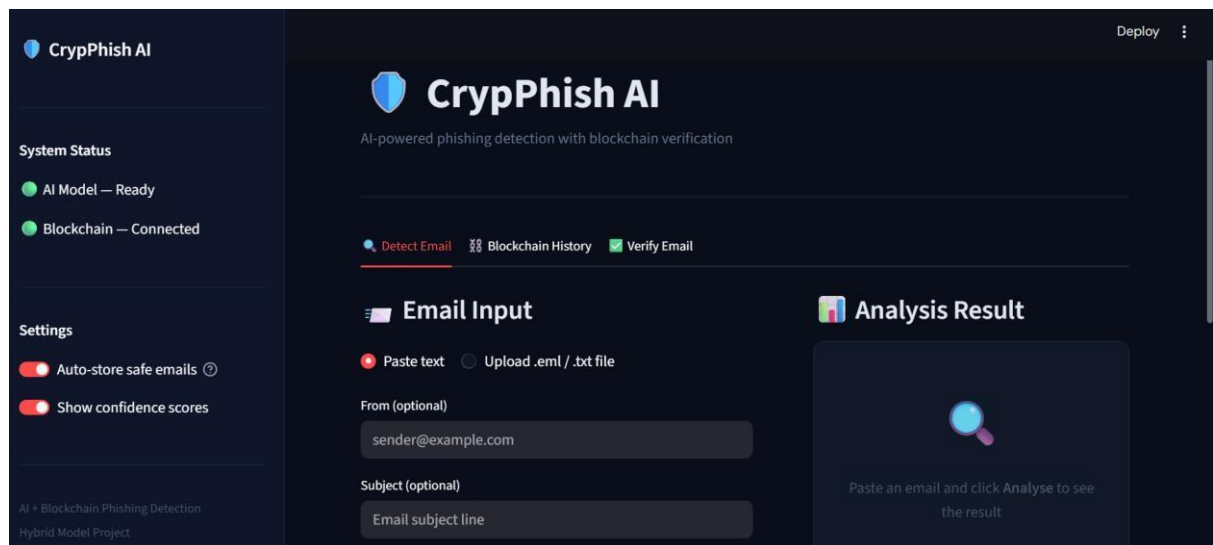


Figure 23: Web Application Interface for the system

CrypPhish AI is designed with a web-based interface to ensure that its functionalities in detecting phishing attacks and verifying email security are easy and convenient for the end-user.

The interface is segregated into various parts for easier interaction from the end-user's perspective. The left panel of the interface shows the system status, which tells whether the AI algorithm is up and running or not. Similarly, it also shows whether the blockchain network is connected or not.

The core part of the interface consists of three major modules, namely Detect Email, Blockchain History, and Verify Email.

The Detect Email module will allow users to enter information on the email either through pasting or importing an email file in .eml or .txt format. There are other parameters like 'Sender', 'Subject' and 'Body' to aid in the process of extraction.

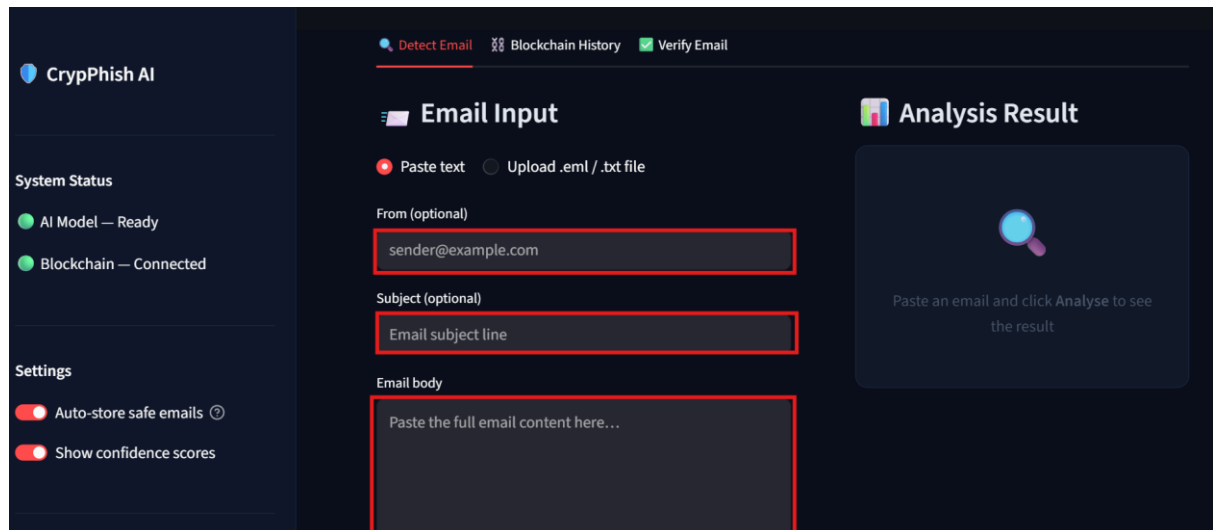


Figure 24: Various features of the Web Application Interface

The Analysis Result module shows the output of the analysis done by the machine learning algorithm. This involves classifying the email as phishing or non-phishing.

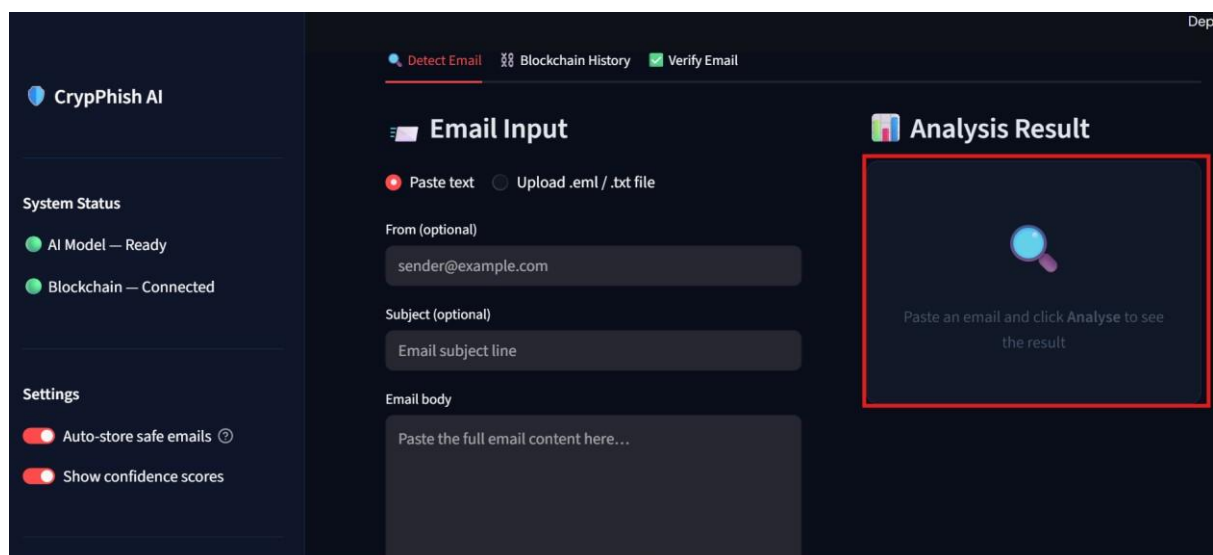


Figure 25: Highlight of the Result Analyse feature of the Web Application

The Blockchain History module lets users retrieve previous records on the block chain for the emails stored on the system.

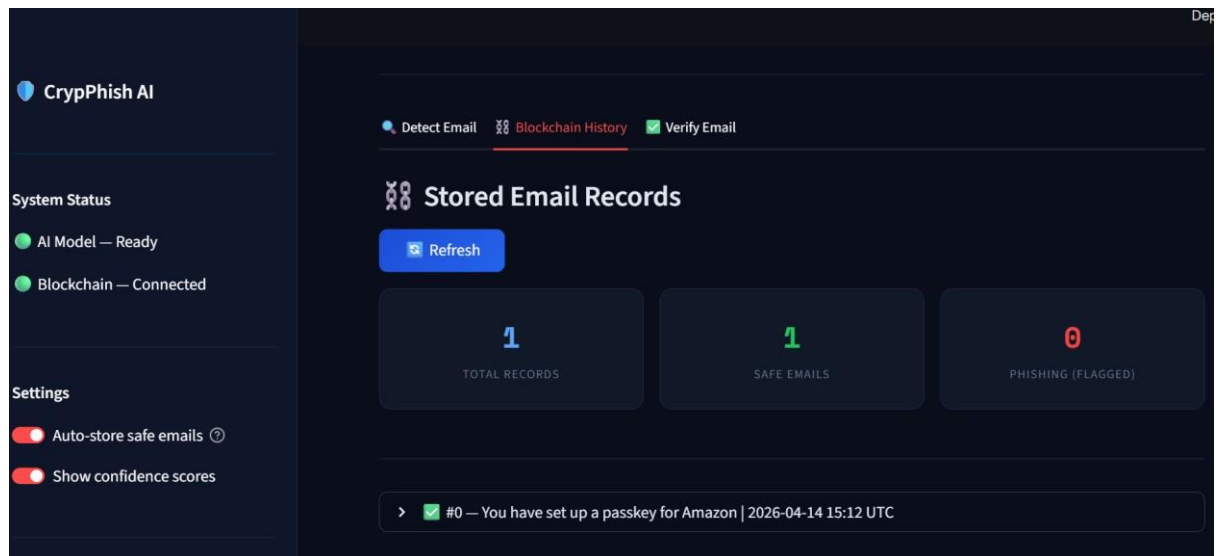


Figure 26: Highlight of the Blockchain History feature

The Verify Email module will help users verify the authenticity of emails using the hashed value recorded on the block chain.

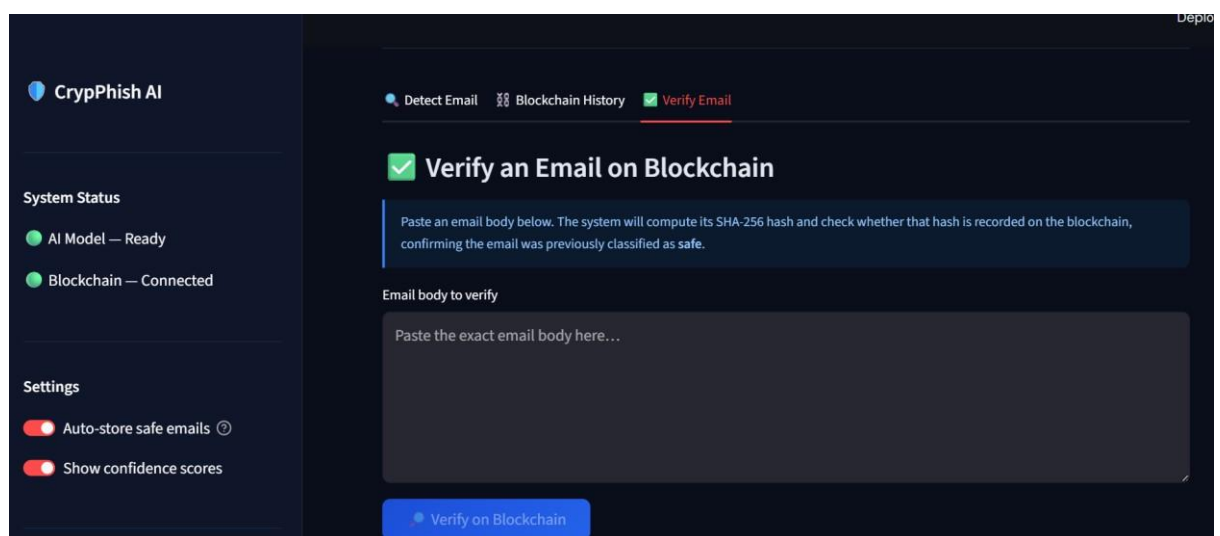


Figure 27: Highlight of the Verify Email feature of the Web Application

There are several settings that can be configured in the application to enable users to tailor it to their needs, like storing all legitimate emails automatically.

On the whole, the web interface is an example of practical application of the suggested system with machine learning based detections and blockchain technology verification being used together.

4.8 EXAMPLE OF SYSTEM OUTPUT ANALYSIS

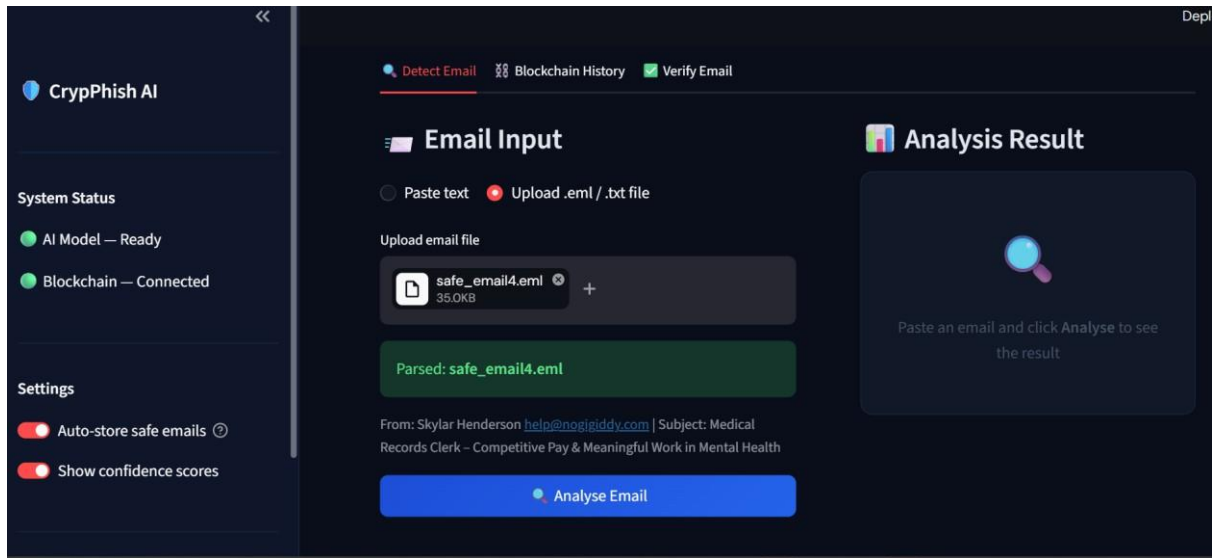


Figure 28: Email Sample being tested on the Web Application

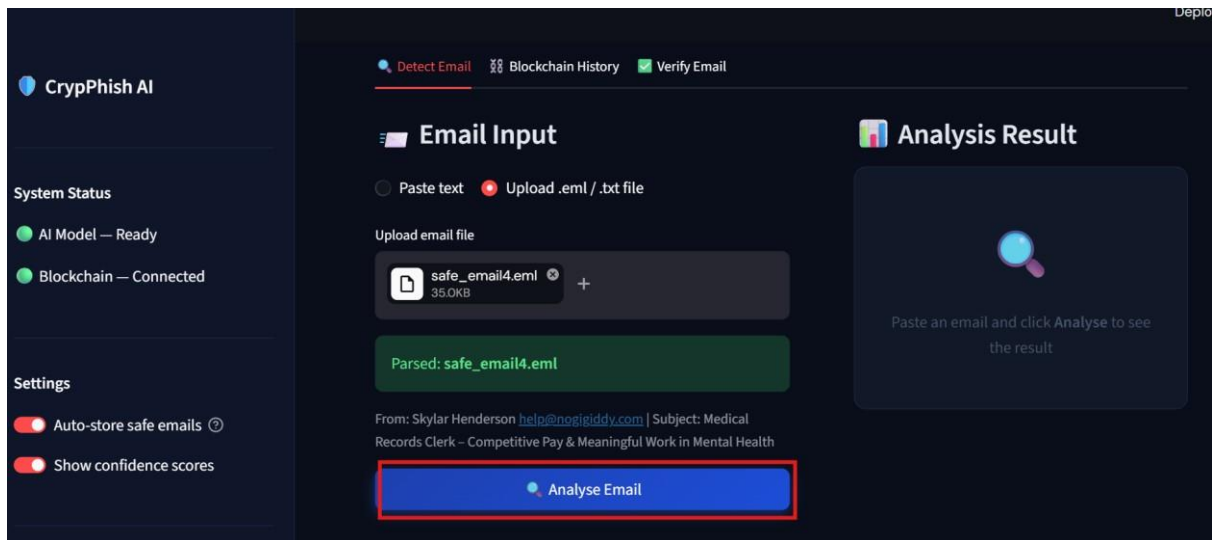


Figure 29: Email ready to be analysed

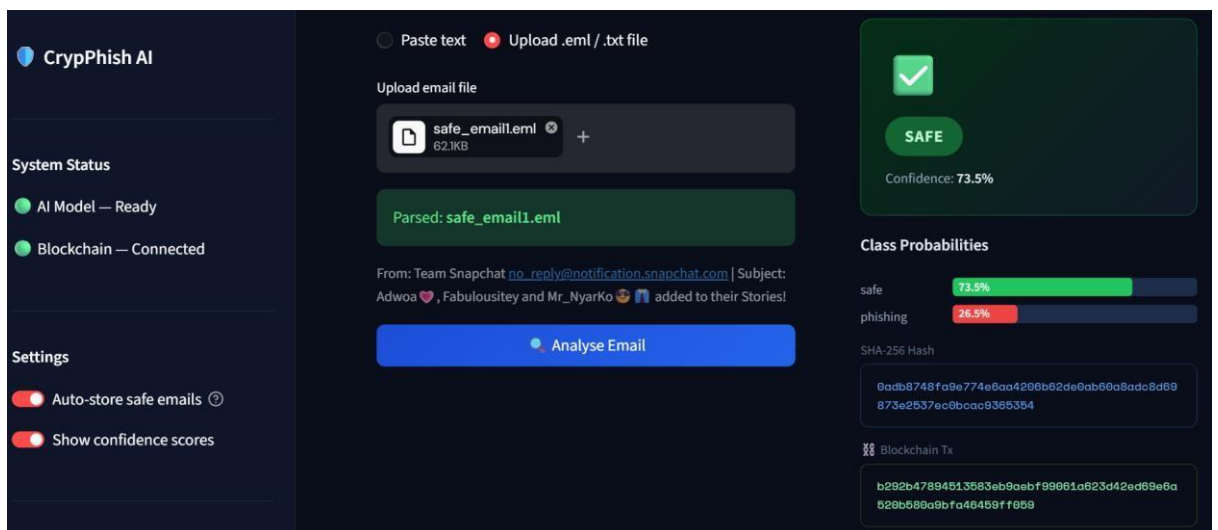


Figure 30: Example Output of CrypPhish AI System Showing Safe Email Classification and Blockchain Verification

This is the illustration of the output sample produced by the system on the email file through CrypPhish AI interface.

From the illustrations, it can be observed that the user uploads the email file named `safe_email1.eml`. The system is able to parse the email file and display the metadata (sender and subject).

After analyzing this particular email file, the system categorizes it as "SAFE". The system models provide a confidence score of 73.5% for this particular categorization. Apart from this, the system models provide the distribution of classes in terms of probability as: 73.5% SAFE and 26.5% PHISHING.

Furthermore, a hash of SHA-256 type is created from the email content, and this functions as a digital signature. The hash thus created is stored on the blockchain, as evident from the blockchain transaction (Tx) feature available in the system. In other words, this ensures that the data related to the email has been secured.

The integration of classification results and validation through blockchain in one interface is indicative of how efficient the system is. With the machine learning algorithm used by the system being able to detect phishing attempts in real-time, the safety and integrity of the emails verified using blockchain have been ensured.

This scenario shows how functional the system can be and how well it combines intelligent detection and secure validation.

4.9 EXAMPLE OF PHISHING DETECTION OUTPUT

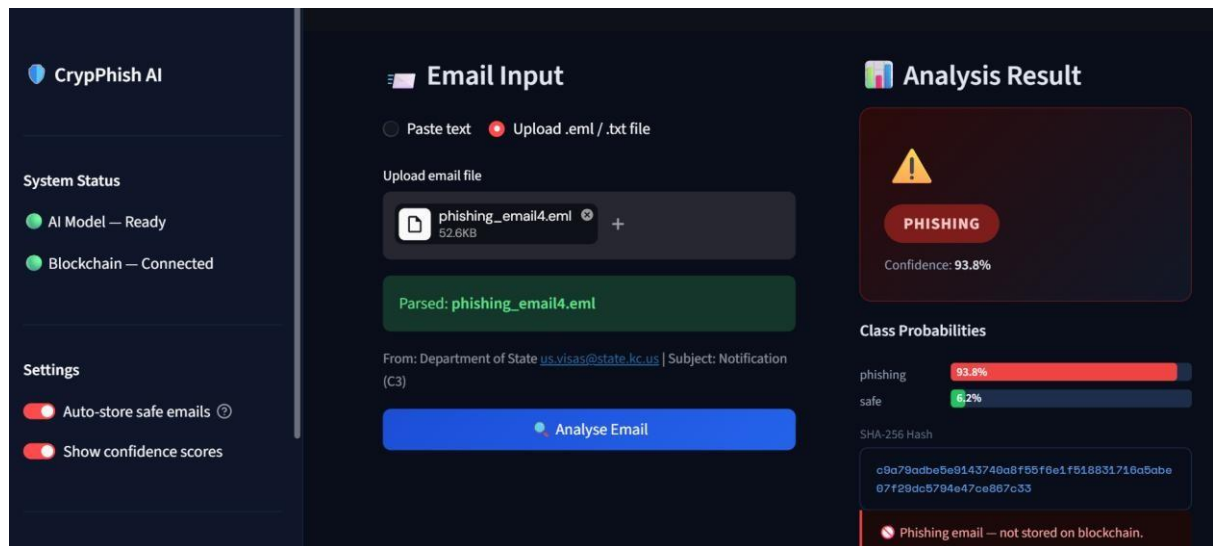


Figure 31: Example Output Showing Phishing Email Detection with High Confidence

The diagram shown below represents one such instance where an input to the system was analysed using the CrypPhish artificial intelligence system.

Here, an email file called "phishing_email4.eml" is uploaded by the user. This file is processed successfully and its metadata, including information about the sender and subject matter, is presented to provide relevant context regarding the classification process.

Upon analysing this email, the system is able to determine that this is indeed a phishing email with a confidence score of 93.8%. This can be corroborated from the probability distribution between classes.

The system does not use the blockchain to store phishing emails. This fact is explicitly shown in the interface, as a statement says that this phishing email is not registered in the blockchain. The reason behind such a design is to make the storage optimized and ensure that the email is stored only if it is legitimate and verified.

Also, even though the email is not registered, a SHA-256 hash can be created for it and may be used later if necessary. The fact that there is no storage of phishing emails in the blockchain further shows the selective nature of the storage performed by the system.

It can be seen that the proposed solution has worked effectively and managed to detect phishing emails without allowing any malicious information to be stored in the blockchain.

4.10 OVERALL DISCUSSION AND CONCLUSION OF RESULTS

The outcomes that are described in this chapter illustrate the efficiency of the developed hybrid phishing detection solution, called CrypPhish AI, which combines machine learning and blockchain into one framework.

In terms of machine learning, it can be noted that the models such as SVM, random forest, and logistic regression demonstrate a fairly high degree of efficiency in the detection of phishing in email messages. Specifically, SVM is shown to be the most efficient algorithm considering all criteria under study. Furthermore, according to the received findings, classical ML approaches demonstrate higher effectiveness than deep learning algorithms, including LSTM, due to the characteristics of the dataset.

The implementation of the blockchain technology is a very important factor from the viewpoint of security because thanks to this technology, the system becomes quite reliable and credible. To begin with, the technology of SHA-256 and smart contracts provides the safety of storing emails. Another aspect that should be taken into account in relation to the blockchain component of the app concerns traceability and transparency.

Additionally, it can be noted that the developed web interface clearly indicates the practicality of the project. Being an interactive tool, through which users can input a new email address and receive the results of its detection and verification through a blockchain network, the web interface demonstrates the applicability of theoretical research in practical terms.

Overall, the incorporation of machine learning technology along with the blockchain technologies creates a powerful tool capable of solving problems related to both accuracy and security of email analysis.

4.11 COMPARISON WITH EXISTING TECHNIQUES

There have been many studies carried out on how to detect phishing scams through various techniques, which include conventional machine learning, cryptology-based, and even the use of blockchain. All these techniques have their strengths and weaknesses.

Conventional machine learning has proven to be efficient when used to detect phishing emails through pattern recognition. Nevertheless, conventional machine learning does not offer any means of secure storage and validation of the recognized data. Therefore, it lacks trust and integrity features.

Security enhancement through cryptography-based systems works by providing authentication and confidentiality through SPF, DKIM, and DMARC protocols. Even though these methodologies are

helpful in verifying the authenticity of emails, they do not help much when it comes to dealing with phishing attacks that are conducted on compromised legitimate domains.

Decentralization, immutable, and transparent nature is what a typical blockchain-based system brings into play. Many of the available systems either lack the required detection abilities or have high computational cost.

In order to address all these drawbacks, the proposed framework uses both machine learning and blockchain technologies together, and that is how CrypPhish AI framework works.

Table III : Comparison of Existing Methods with Proposed System

Method	Detection Accuracy	Security	Data Integrity	Real-time Capability	Limitations
Machine Learning Only	High	Low	Low	High	No secure storage
CryptographyBased	Medium	High	High	High	Limited detection capability
Blockchain-Based Only	Medium	High	High	Medium	High overhead, weak detection
Proposed System (CrypPhish AI)	High	High	High	High	Computational overhead (manageable)

4.12 DISCUSSION OF COMPARISON

As we see from the above comparison, the proposed solution gives a well-rounded result. Even though classic machine learning algorithms give high accuracy of detection, they do not have security and trust factors. The solution based on cryptography improves authentication features but is not effective in detecting phishing scams. On the other hand, blockchain technologies provide good data integrity but lack detection abilities.

However, combining two solutions into one hybrid system allows integrating advantages of both machine learning and blockchain systems. In particular, the system can detect frauds accurately while giving a possibility to keep all emails stored securely, immutably, and verified.

CHAPTER 5

FUTURE SCOPE

The suggested system, namely CrypPhish AI, shows excellent results in detecting phishing attacks and authenticating emails using the combination of machine learning algorithms and blockchain. Nevertheless, there are still some ways to improve and extend the functionality of this system.

First, there is an option to use more advanced approaches in machine learning, which include methods belonging to the domain of deep learning (such as Transformer architectures, for instance, BERT model). The advantage of deep learning models lies in the fact that they can detect more implicit relationships in data.

The other important feature is the use of large amounts of data for the training process. The use of different datasets for phishing emails will help improve the efficiency of the algorithm.

With regard to blockchain technology, the present system is implemented locally to evaluate its performance. In future studies, there could be an attempt to implement the system using public or consortium blockchains so as to enhance its scalability and practicality. There could also be efforts to reduce transaction costs and optimize transaction times.

The system can be further extended to conduct phishing attacks in real time by integrating the system with live e-mail servers, enabling automated detection of phishing scams in a real-life scenario.

There are other areas where the system can be improved, for example, by incorporating functionalities like real-time warnings, user authentication, and role-based permissions in the web interface. These improvements will not only improve the security of the system but also enhance its usability.

Lastly, the suggested solution may be generalized to detect and prevent other kinds of cyberattacks, including spam, malware, and even social engineering attacks.

CHAPTER 6

CONCLUSION

The CrypPhish AI technology was developed as a result of this study as a novel solution for detecting phishing attacks by employing machine learning algorithms along with blockchain technologies in order to address the increasing threats of phishing attacks in the current digitized world.

The initial step undertaken within the scope of this study was that of identifying the limitations of existing solutions regarding phishing attack detection. Although the techniques based solely on machine learning algorithms are very efficient in the detection of such attacks, they lack the feature of validating their accuracy.

To distinguish between phishing emails, spam, and normal emails, a framework that employs machine learning has been designed. Multiple approaches were examined, such as logistic regression, random forest, SVM, and LSTM. The test outcomes indicate that out of the models tested, SVM demonstrated maximum accuracy for email detection using all parameters: accuracy, precision, recall, and F1 score. Furthermore, tests have demonstrated that traditional machine learning algorithms outperform deep learning approaches when applied to a specific dataset since they yield better results on structured datasets.

Notably, blockchain technology is applied not only for phishing email detection but also as a means of securing data. In particular, the system uses SHA-256 hashing algorithm and smart contracts to create immutable and unalterable records of valid emails on the blockchain. This provides transparency, traceability, and reliability. Moreover, emails are selected selectively for the recording process in order to optimize the usage of computing resources.

To make the model more interactive and user-friendly, a user interface using a graphical interface that runs on the web platform was designed. Using this platform, a user can easily send emails, identify phishing emails, view the confidence score, and verify the emails with block chain technology.

In conclusion, it should be noted that the combination of machine learning and blockchain technologies helped to develop a comprehensive framework for detecting phishing as it addressed such issues as detection effectiveness and secure messaging. Therefore, on one hand, the system improves the ability of detecting phishing messages while, on the other hand, it was also aimed at providing safe communication channels.

Overall, the discussed research work made a good contribution to cybersecurity as the developed framework includes highly efficient tools for solving such problem as phishing.

REFERENCES

- [1] A. Khan, M. Ahmed, and A. Fathima, "Enhanced phishing detection using machine learning algorithms: A comparative study of random forest, svm, and logistic regression models," SSRN Electronic Journal, 2025, preprint. [Online] Available: <https://ssrn.com/abstract=5191566>
- [2] B. Lim, R. Huerta, A. Sotelo, A. Quintela, and P. Kumar, "Explicate: Enhancing phishing detection through explainable ai and llm-powered interpretability," arXiv preprint arXiv:2503.20796, 2025.
- [3] P. C. R. Chinta, C. S. Moore, L. M. Karaka, M. Sakuru, V. Bodepudi, and S. R. Maka, "Building an intelligent phishing email detection system using machine learning and feature engineering," European Journal of Applied Science, Engineering and Technology, vol. 3, no. 2, pp. 41–54, March–April 2025.
- [4] E. O. Asani, V. O. Adedayo-Ajayi, A. E. Tunbosun, N. Enumah, and D. R. Aremu, "Detection of phishing emails using support vector classifier and gaussian latent variable model," in 2024 International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG). Omu Aran, Nigeria: IEEE, 2024.
- [5] S. S, V. S, C. M, and V. R. A, "Ai sentries: Evaluating machine learning models for superior phishing email detection," in 2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS). Bengaluru, India: IEEE, 2024.
- [6] S. Mahalakshmi, D. L. Pansy, and V. M. Thejashree, "Smart email filtering against phishing attacks," in 2025 International Conference on Data Science, Agents, and Artificial Intelligence (ICDSA AI). Chennai, India: IEEE, 2025.
- [7] P. Zhao and S. Jin, "Fewshing: A few-shot learning approach to phishing email detection," in 2024 IEEE 4th International Conference on Software Engineering and Artificial Intelligence (SEAI). Guangzhou, China: IEEE, 2024.
- [8] I. Vasudevan, R. M. D. Charaan, D. Amirtharaj, S. A. Kumar, A. Jaisuriya, and L. Jeneffa, "Hybrid model for email spam detection using multi-modal knn and svm," in 2024 International Conference on Smart Technologies for Sustainable Development Goals (ICSTSDG). IEEE,

- [9] F. Heiding, B. Schneier, A. Vishwanath, J. Bernstein, and P. S. Park, “Devising and Detecting Phishing Emails Using Large Language Models,” *IEEE Access*, vol. 12, pp. 42131–42146, 2024, doi: 10.1109/ACCESS.2024.3375882.
- [10] P. T. Duy, V. Q. Minh, B. T. H. Dang, N. D. H. Son, N. H. Quyen, and V.-H. Pham, “A Study on Adversarial Sample Resistance and Defense Mechanism for Multimodal Learning-Based Phishing Website Detection,” *IEEE Access*, vol. 12, pp. 137805–137824, 2024, doi: 10.1109/ACCESS.2024.3436812.
- [11] D. Jennifer Dsouza, A. P. Rodrigues, and R. Fernandes, “Multi-Modal Comparative Analysis on Execution of Phishing Detection Using Artificial Intelligence,” *IEEE Access*, vol. 12, pp. 163016–163041, 2024, doi: 10.1109/ACCESS.2024.3491429.
- [12] O. K. Sahingoz, E. BUBEr, and E. Kugu, “DEPHIDES: Deep Learning Based Phishing Detection System,” *IEEE Access*, vol. 12, pp. 8052–8070, 2024, doi: 10.1109/ACCESS.2024.3352629.
- [13] P. Zhao and S. Jin, “Fewshing: A Few-Shot Learning Approach to Phishing Email Detection,” in *2024 IEEE 4th International Conference on Software Engineering and Artificial Intelligence (SEAI)*, Xiamen, China: IEEE, Jun. 2024, pp. 371–375. doi: 10.1109/SEAI62072.2024.10674290.
- [14] L. Jovanovic et al., “Improving Phishing Website Detection Using a Hybrid Two-level Framework for Feature Selection and XGBoost Tuning,” *J. Web Eng.*, Jul. 2023, doi: 10.13052/jwe1540-9589.2237.
- [15] A. Khalid, M. Hanif, A. Hameed, Z. Ashraf, M. M. Alnfai, and S. M. M. Alnefaie, “LogiTriBlend: A Novel Hybrid Stacking Approach for Enhanced Phishing Email Detection Using ML Models and Vectorization Approach,” *IEEE Access*, vol. 12, pp. 193807–193821, 2024, doi: 10.1109/ACCESS.2024.3518923.
- [16] S. Asiri, Y. Xiao, S. Alzahrani, S. Li, and T. Li, “A Survey of Intelligent Detection Designs of HTML URL Phishing Attacks,” *IEEE Access*, vol. 11, pp. 6421–6443, 2023, doi: 10.1109/ACCESS.2023.3237798. S. Asiri, Y. Xiao, S. Alzahrani, S. Li, and T. Li, “A Survey of Intelligent Detection Designs of HTML URL Phishing Attacks,” *IEEE Access*, vol. 11, pp. 6421–6443, 2023, doi: 10.1109/ACCESS.2023.3237798.
- [17] D. He, Z. Liu, X. Lv, S. Chan, and M. Guizani, “On Phishing URL Detection Using Feature Extension,” *IEEE Internet Things J.*, vol. 11, no. 24, pp. 39527–39536, Dec. 2024, doi: 10.1109/JIOT.2024.3446894.

- [18] A. Karim, M. Shahroz, K. Mustofa, S. B. Belhaouari, and S. R. K. Joga, "Phishing Detection System Through Hybrid Machine Learning Based on URL," *IEEE Access*, vol. 11, pp. 36805–36822, 2023, doi: 10.1109/ACCESS.2023.3252366.
- [19] L. R. Kalabarige, R. S. Rao, A. R. Pais, and L. A. Gabralla, "A Boosting Based Hybrid Feature Selection and Multi-Layer Stacked Ensemble Learning Model to Detect Phishing Websites," *IEEE Access*, vol. 11, pp. 71180–71193, 2023, doi: 10.1109/ACCESS.2023.3293649.
- [20] J. Luo, J. Qin, R. Wang, and L. Li, "A Phishing Account Detection Model via Network Embedding for Ethereum," *IEEE Trans. Circuits Syst. II Express Briefs*, vol. 71, no. 2, pp. 622–626, Feb. 2024, doi: 10.1109/TCSII.2023.3267822.
- [21] G. Brezeanu, A. Archip, and C.-G. Artene, "Phish Fighter: Self Updating Machine Learning Shield Against Phishing Kits Based on HTML Code Analysis," *IEEE Access*, vol. 13, pp. 4460–4486, 2025, doi: 10.1109/ACCESS.2025.3525998.
- [22] F. Castaño, E. F. Fernández, R. Alaiz-Rodríguez, and E. Alegre, "PhiKitA: Phishing Kit Attacks Dataset for Phishing Websites Identification," *IEEE Access*, vol. 11, pp. 40779–40789, 2023, doi: 10.1109/ACCESS.2023.3268027.
- [23] B. B. Gupta, A. Gaurav, and K. T. Chui, "Optimized deep learning model for phishing detection in blockchain transactions using bert and teaching learning-based algorithm," in *2024 IEEE Future Networks World Forum (FNWF)*. Taichung, Taiwan: IEEE, 2024, pp. 765–770.
- [24] A. Rachad, L. Gaiz, K. Bouragba, and M. Ouzzif, "Sending and receiving secure email based on blockchain," in *2024 International Conference on Ubiquitous Networking (UNet)*. Casablanca, Morocco: IEEE, 2024.
- [25] J. N. Al-Karaki, A. Gawanmeh, and C. Fachkha, "Blockchain for email security: A perspective on existing and potential solutions for phishing attacks," in *2023 Fifth International Conference on Blockchain Computing and Applications (BCCA)*. Abu Dhabi, UAE: IEEE, 2023, pp. 404–411.
- [26] Y. Xia, J. Liu, and J. Wu, "Phishing detection on ethereum via attributed ego-graph embedding," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 69, no. 5, pp. 2538–2540, 2022.
- [27] A. Ray, S. Sakunthala, and A. Prabhakar, "Improving phishing detection in ethereum transaction network using quantum machine learning," in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*. USA: IEEE, 2023, pp. 1107–1113.

- [28] M. B. Hossain, M. Rahayu, M. A. Ali, S. Huda, Y. Koderu, and Y. Nogami, "A smart contract based blockchain approach integrated with elliptic curve cryptography for secure email application," in 2023 Eleventh International Symposium on Computing and Networking Workshops (CANDARW). Okayama, Japan: IEEE, 2023, pp. 195–201.
- [29] M. P. R, S. A. C. D, and S. P, "Safeblock mail– enhancing email security using block chain technology," in 2024 International Conference on Emerging Research in Computational Science (ICERCS). Chennai, India: IEEE, 2024.
- [30] C. Xu, R. Li, L. Zhu, K. Sharif, and X. Shen, "Ewdps: A novel framework for early warning and detection on ethereum phishing scams," IEEE Internet of Things Journal, vol. 11, no. 19, pp. 30483–30494, October 2024.
- [31] J. Kang and S.-J. Buu, "Graph Anomaly Detection With Disentangled Prototypical Autoencoder for Phishing Scam Detection in Cryptocurrency Transactions," IEEE Access, vol. 12, pp. 91075–91088, 2024, doi: 10.1109/ACCESS.2024.3419152.

PUBLICATIONS

“A Hybrid Approach for Phishing Email Detection Using AI and Cryptography,” in *Proc. 5th Int. Conf. Pervasive Computing and Social Networking (ICPCSN)*, Chennai, India, 2025(Published).

“A Hybrid Framework for Phishing Email Detection and Secure Communication Using AI and Blockchain,” in *Proc. IEEE 7th Int. Conf. Computing, Communication and Automation (ICCCA)*, Noida, India, 2025 (Published).